

COMMUTATIVE ALGEBRA

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1. RINGS

Definition 1.1. A ring R is called a *domain* if it has no proper zero divisors, that is, if $ab = 0$ then $a = 0$ or $b = 0$.

Theorem 1.2. *Let R be a ring and $I \subset R$ an ideal. There is a correspondence between ideals of R containing I and ideals of R/I .*

Proof. We show that there are two maps between the set of ideals of R containing I and the ideals of R/I whose compositions are the identities in the corresponding set. One map is the projection to the quotient, and the other map is taking the elements whose equivalence class lies in a given ideal.

Suppose that $\tilde{J}$ is an ideal of R/I . Define $J = \{j \in R : j + I \in \tilde{J}\}$, the set of elements that project to $\tilde{J}$. Then $I \subset J$, because $i + I = I$ is the zero of R/I which is contained in any ideal of R/I . Also notice that J is an ideal of R since for any $r \in R$, $rj + I \in \tilde{J}$ by $\tilde{J}$ being an ideal. Notice that projecting J back to R/I gives $\tilde{J}$ by definition.

Conversely, we project to the quotient. Let J be any ideal of R containing I . Let $\tilde{J}$ be the projection to R/I . Then $\tilde{J}$ is an ideal of R/I by J being an ideal of R . Consider the set J' of all elements whose projection lies in $\tilde{J}$.

Then $J' = J$ by definition. $\square$

This is the essential tool in proving that R/I is a field when I is a maximal ideal

Lemma 1.3. *Let R be a ring and I a maximal ideal of R . Then R/I is a field.*

Proof. It's easy to prove that a ring is a field if and only if its only ideals are $\{0\}$ and R . By Theorem 1.2, the ideals of R/I are in correspondence with ideals of R containing I , but only R is such. $\square$

2. POLYNOMIAL RINGS

Lemma 2.1. *The ring of polynomials in one indeterminate over a domain R is a domain.*

Proof. Suppose that $fg = 0$ for $f, g \in R[x]$. Then the leading coefficient of fg is a_nb_m where $f = \sum_{i=1}^n a_ix^i$ and $g = \sum_{i=1}^m b_ix^i$. Since $a_nb_m = 0$, then $a_n = 0$ or $b_m = 0$, but this is a contradiction since R is a domain and $a_n \neq 0$, $b_m \neq 0$. $\square$

3. GREATEST COMMON DIVISOR AND LEAST COMMON MULTIPLE

Following is the universal definition of lcm and gcd:

Definition 3.1. A *least common multiple* of $a, b \in R$ is an element d such that if for every $c \in R$, if $a, b \mid c$ then $d \mid c$.

A *greatest common divisor* of $a, b \in R$ is an element d such that for every $c \in R$, if $c \mid a, b$ then $c \mid d$.

But it looks like it shall be easier to define them as follows:

Definition 3.2. A *least common multiple* of $f, g \in R[x]$ is the monic polynomial d of smallest degree such that $f, g \mid d$.

A *greatest common divisor* of $f, g \in R[x]$ is the monic polynomial d of greatest degree such that $d \mid f, g$.

Lemma 3.3. *If R is a UFD, in $R[x]$ there exist lcm and gcd, and are unique up to multiplication by units.*

Proof. We can order by degree, and those of the same degree order by the leading term... but this requires an order in R . $\square$

4. EUCLIDEAN DOMAINS

Definition 4.1. A domain R is an *Euclidean domain* if there is a function $\varphi : R \rightarrow \mathbb{Z}_+$, the positive integers, such that

- (1) $\varphi(ab) \geq \varphi(a)$ for all $a, b \in R$,
- (2) For all $a, b \in R$ with $\varphi(a) \geq \varphi(b)$, there are $q, r \in R$ such that

$$a = bq + r \quad \text{and} \quad \varphi(r) < \varphi(b).$$

Remark 4.2. Item 1 is equivalent to asking that if a divides b then $\varphi(a) \leq \varphi(b)$.

Theorem 4.3 (Euclidean algorithm for polynomial rings). *Let R be a domain and $f, g \in R[x]$ with the leading coefficient of g a unit and $\deg(f) \geq \deg(g)$. Then there are polynomials $q, r \in R[x]$ such that $f = qg + r$ with $\deg(r) < \deg g$.*

Proof. Since $\deg(r)$ must be smaller than $\deg q$ and the leading coefficient of a product is the product of the leading coefficients of the factors, we readily see that the leading coefficient of q must be $L(f)/L(g)$ and of degree $n - m$, where L is the leading coefficient function and $\deg(f) = n$, $\deg(g) = m$.

We are lead to the following natural strategy: write $g = L(g)x^m + g_1$ for some g_1 of degree strictly smaller than m . Likewise, write $f = L(f)x^n + f_1$.

Then multiply g with the leading term of q to obtain

$$\begin{aligned} g \left(\frac{L(f)}{L(g)} x^{n-m} \right) &= (L(g)x^m + g_1) \left(\frac{L(f)}{L(g)} x^{n-m} \right) \\ &= L(f)x^n + g_1 \left(\frac{L(f)}{L(g)} x^{n-m} \right). \end{aligned}$$

Add f_1 on both sides to obtain f :

$$\begin{aligned} g \left(\frac{L(f)}{L(g)} x^{n-m} \right) + f_1 &= L(f)x^n + f_1 + g_1 \left(\frac{L(f)}{L(g)} x^{n-m} \right) \\ \iff f &= g \left(\frac{L(f)}{L(g)} x^{n-m} \right) + f_1 - g_1 \left(\frac{L(f)}{L(g)} x^{n-m} \right) \end{aligned}$$

We have great candidates for q and r , except that put this way it is not necessarily true that the remainder will have degree strictly smaller than g . However, it is true that it will have degree strictly smaller than the degree of f . Then we can just repeat the process until we arrive at a reminder with degree strictly smaller than the degree of g . $\square$

Exercise 4.4. Prove that the ring of Gaussian integers $\mathbb{Z}[\sqrt{-1}]$ is Euclidean.

Proof. Take two Gaussian integers p, q and take their quotient in $\mathbb{C}$. By simple geometrical observation, the nearest Gaussian integer s is at distance at most $1/\sqrt{2}$ from p/q . That is,

$$\left| \frac{p}{q} - s \right| < \frac{1}{\sqrt{2}}.$$

Multiplying by $|q|$ we obtain

$$|p - qs| \leq \frac{|q|}{\sqrt{2}}.$$

Indeed — we can view the squared norms of Gaussian integers as $x\bar{x}$, so that

$$|q|^2 \left| \frac{p}{q} - s \right|^2 = q\bar{q} \left(\frac{p}{q} - s \right) \overline{\left(\frac{p}{q} - s \right)} = \left(\frac{p}{q} - qs \right) \overline{\left(\frac{p}{q} - qs \right)} = |p - qs|^2.$$

Defining $r = p - qs$ yields the result. $\square$

The same argument shows that $\mathbb{Z}$, $\mathbb{Z}[\sqrt{-2}]$ and $\mathbb{Z} \left[\frac{1+\sqrt{-3}}{2} \right]$ are also Euclidean rings. But we can't use it for $\mathbb{Z}[\sqrt{-3}]$ or $\mathbb{Z}[\sqrt{5}]$ — is any of these Euclidean? Neither, since they are not integrally closed, while we have:

Lemma 4.5. *A PID is integrally closed.*

Proof. ? $\square$

5. UNIQUE FACTORIZATION DOMAINS AND THE GAUSS LEMMA

The point of the Gauss lemma is that a polynomial f over the field of fractions F of a UFD R can be expressed as a rational number called the content of f times a polynomial over R called the primitive part. The point of the Gauss lemma is that it allows us to make sure that the content will in fact be an element of R .

The following lemma shows that in UFDs irreducibles are primes, but this is not true in general. (Counterexample in $\mathbb{Z}[\sqrt{-5}]$.)

Lemma 5.1. *If R is a UFD then irreducible elements are prime.*

Proof. If f is irreducible and $f|ab$ then f must be an element of the unique factorization of a or b . $\square$

In fact, showing that irreducibles are prime would be enough for our purpose of showing that $F[x]$ is a UFD when F is a field. We shall be able to prove this (without the assumption of UFD as above) via Bachet-Bézout theorem; which we state first in its version for integer numbers:

Theorem 5.2 (Bachet-Bézout for integers). *Let $a, b \in \mathbb{Z}$ (what is the essential property of $\mathbb{Z}$ that makes this true?). Then there exist $x, y \in \mathbb{Z}$ with*

$$ax + by = \gcd(a, b)$$

Therefore, if $c \in \mathbb{Z}$ divides both a and b then c divides $\gcd(a, b)$ as well.

Lemma 5.3. *Let $a, b, c \in \mathbb{Z}$. If $a|bc$ and $(a, b) = 1$, then $a|c$.*

Proof. By Theorem 5.2, there exist elements p, q such that $ap + bq = 1 \implies apc + bcq = c$, so that a divides both summands and thus divides c . $\square$

Theorem 5.4 (Division algorithm for polynomials). *Let F be a field. Given two polynomials $f, g \in F[x]$ there are two unique $q, r \in F[x]$ such that*

$$f = qg + r, \quad \deg r < \deg g.$$

Remark. *We use the hypothesis that F is a field to take a fraction with denominator the leading coefficient of g ; that is, the theorem is valid if the leading coefficient of g is a unit.*

Proof. We may suppose that $\deg f \geq \deg g$ for if not we just pick $q = 0$ and $r = f$. Now we use induction. If $\deg f = \deg g = 0$ then f and g are numbers and the result is obvious since F is a field.

If $\deg f > 0$ we do as follows. Suppose $\deg f = n$ and $\deg g = m$. Write $f(x) = a_n x^n + f_1(x)$ and $g(x) = b_m x^m + g_1(x)$. Notice that

$$\frac{a_n x^n}{b_m x^m} g(x) = a_n x^n + \frac{a_n}{b_m} x^{n-m} g_1(x)$$

so that $f(x) - \frac{a_n}{b_m} x^{n-m} g(x) = f_1(x) - \frac{a_n}{b_m} x^{n-m} g_1(x)$. This is a polynomial of strictly less degree than that of f , so that we can use induction hypothesis to write it as $qg + r$ with $\deg r < m$. Then

$$f(x) = g(x) \left(q(x) + \frac{a_n}{b_m} x^{n-m} \right) + r(x).$$

To prove uniqueness suppose that $f = q_1 g + r_1 = q_2 g + r_2$. Then

$$(q_1 - q_2)g = r_1 - r_2$$

Taking degrees we see that the degree of this polynomial should be, by the right-hand-side less than the degree of g , and by the left hand side more than the degree of g . Then the degree must be zero and $q_1 = q_2$, $r_1 = r_2$. $\square$

Recall that the greatest common divisor of two polynomials f and g is the *monic* polynomial of least degree that divides both f and g .

Theorem 5.5 (Bachet-Bezout for polynomials over field). *Let F be a field and $f, g \in F[x]$. There exist elements $p, q \in F[x]$ such that $fp + gq = \gcd(f, g)$.*

Proof. Consider the set

$$I := \{fx + gy : x, y \in F[x]\}.$$

Let $d = fp + gq$ be the monic polynomial of least degree of the form $fx + gy$, for $x, y \in F[x]$. Notice that the existence of d monic comes from F being a field. We claim that $d = \gcd(f, g)$. Indeed, we first check by division algorithm that d divides both f and g : suppose that $f = dq + r$ with $\deg r < \deg d$. Then, if $r = f - dq = f(1 - xq) - yg$ is not zero, dividing by its leading coefficient gives an element in I of degree strictly smaller than the degree of d , a contradiction. Thus r is zero. The same argument shows that $d|g$, and we conclude that $\deg d \leq \deg \gcd(f, g)$.

Now, since $\gcd(f, g)$ divides both f and g , it divides $d = xf + yg$. (Indeed, if $c := \gcd(f, g)$ is such that $ca = f, cb = g$ then $c(ap + bq) = cap + cbq = fp + gq = d$.) Thus $\deg d = \deg \gcd(f, g)$, and since both are monic we are done. $\square$

Finally we obtain

Lemma 5.6. *Let F be a field. Irreducible elements of $F[x]$ are prime. That is, if $f|gh$ then $f|g$ or $f|h$ for any $f, g, h \in F[x]$.*

Proof. Let $d = \gcd(f, g)$. Then $dp = f$ and $dq = g$. If d is not 1 we get that p is a unit. Then gp is a multiple by a unit of g which is divided by f : $gp = dpq = fq$. That's good enough.

Now suppose that $d = 1$. Then by Bachet-Bezout Theorem 5.5 there are $p, q \in F[x]$ such that $fp + gq = 1$. Then $fph + gqh = h$. Since by hypothesis $f|gh$ we conclude that f divides both summands, so $f|h$. $\square$

Applying this last lemma over and over again we get that if $f|f_1 \dots f_k$ then f must divide f_i for some i .

Lemma 5.7. *If F is a field then $F[x]$ is a UFD.*

Proof. The existence of the factorization is an easy induction on the degree of $f \in R[x]$. If f has degree zero, then we are done. If f has positive degree and is irreducible, we are done. If f has positive degree and it is reducible, i.e. $f = gh$, then both g and h have strictly smaller degree and we may apply induction hypothesis to obtain a decomposition of each factor into irreducible elements.

To prove uniqueness suppose that there exists an element in $F[x]$ that admits more than one factorization in irreducible elements up to multiplication by units. Choose f so that $\deg f$ is minimal among all such elements.

Write two factorizations of f in irreducible elements as $f_1 \dots f_k = f'_1 \dots f'_\ell$. Suppose that the factors are in ascending order of degree.

We have that $f_1|f'_1 \dots f'_{k'}$, so that f_1 must divide some $f'_{i'}$ by Lemma 5.6. Since $f'_{i'}$ is irreducible and f_1 is not a unit, we conclude that $f_{i'} = f_1 u$ for some unit u .

Then we apply cancellation law using that $R[x]$ is a domain to obtain $uf_2 \dots f_k = f'_2 \dots f'_{k'}$ is a polynomial with two distinct factorizations. Since f is minimal among all elements that admit more than one factorization, we conclude that $k = k'$ and $f_i = f'_i$ for all i . $\square$

Remark 5.8. Notice that the proof given for integers does not work identically, for in this case we cannot conclude that $f_1 = f'_1$ directly from putting the factors in order of degree (we can conclude that they have the same degree, but not that they are the same polynomial!).

The true definition of content can be done in three steps:

- Definition 5.9.**
- (1) (Order of a fraction a at a prime p .) Let $a \in F$. If $p \in R$ is irreducible, the *order* of a at p to be the number r such that $a = p^r b$ for some fraction b that does not have p as factor in the numerator nor in the denominator. (To understand the point of the Gauss lemma, notice that the order can be a fraction, i.e. if p is a factor of the denominator.)
 - (2) (Order of a polynomial $f \in F[x]$ at a prime p .) The *order* of $f \in F[x]$ at p is the minimum order of p in any of the coefficients of f . (Defining this number as the **minimum** ensures that the primitive part of f (see below) has all coefficients in R . It also follows that the gcd of the coefficients of the primitive part is 1.)
 - (3) (Content of $f \in F[x]$) The *content* of f is the product of every irreducible factor p of f raised to its order.
 - (4) (Primitive polynomial.) A polynomial is called *primitive* if it has content 1.

As explained in the beginning of this section, the point of the Gauss lemma is that it will allow us to make sure that the content of f is an element of R , i.e. not a fraction.

Also, in the definition of order of a fraction a at a prime p , notice the order well-defined, i.e. unique, by the uniqueness of the factorization in R .

Lemma 5.10 (True Gauss lemma). *Let R be a UFD and F its fraction field. Let $f, g \in F[x]$. Then the $\text{cont}(fg) = \text{cont}(f)\text{cont}(g)$.*

This clearly implies that the product of two primitive polynomials being primitive, and the converse I think is true as well.

Proof. Since $\text{cont}(bf) = b\text{cont}(f)$ it's enough to suppose that f and g are primitive. Indeed, taking content of fg we obtain on one hand $\text{cont}(fg)$ and on the other hand $\text{cont}(\text{cont}(f)\text{pp}(f)\text{cont}(g)\text{pp}(g)) = \text{cont}(f)\text{cont}(g)\text{cont}(\text{pp}(f)\text{pp}(g))$. Thus we would finish if we show that the product of two primitive polynomials is primitive.

Put another way, it's enough to show will show that if f, g are primitive (as are the primitive parts of f and g in the preceding paragraph) then fg is primitive. This means that I will show that any p is not a factor of all the coefficients in fg . A coefficient of fg looks as $c_j = \sum_{i=0}^j a_i b_{j-i}$. Since f, g are primitive then p cannot divide all a_i and all b_i . Thus we can pick the maximum a_s and the maximum b_t that p does not divide. Then the c_j containing the product $a_s b_t$ will not be divided by p , for p cannot divide $a_s b_t$ because it is prime, but will divide the remaining terms in the sum c_j ; thus if p divided c_j it would also divide $a_s b_t$. $\square$

Lemma 5.11 (Gauss). *If R is a UFD then $R[x]$ is a UFD.*

Proof. Let $f \in R[x]$ be distinct from 0. Since $F[x]$ is UFD we can write $f = f_1 \dots f_k$. Notice that the primitive parts of the f_i coincide with the primitive part of the product $f_1 \dots f_k$ by the Gauss Lemma:

$$\begin{aligned} f &= c(f_1 \dots f_k) \text{pp}(f_1 \dots f_k), \\ f &= c(f_1) \text{pp}(f_1) \dots c(f_k) \text{pp}(f_k) \end{aligned}$$

and we cancel the content part dividing (using that we are in a field) by the Gauss Lemma.

We also apply the Gauss Lemma to conclude that $c(f) = c(f_1) \dots c(f_k)$, being the content of a polynomial with coefficients in R , must be in R as well, so that we can decompose it as a product of irreducibles since R is a UFD.

To obtain a factorization in irreducibles we only need to show that each of the $\text{pp}(f_i)$ is irreducible in $R[x]$. To see this suppose that $\text{pp}(f_i) = gh$. $c(f_i) \text{pp}(f_i) = f_i$ is irreducible in $F[x]$, so either $c(f_i)g$ or h is a unit of F . Then either g or h are in R . But the gcd of the coefficients of $\text{pp}(f_i)$ is 1, so, since we have factored an element of R from $\text{pp}(f_i)$, it must be 1.

The uniqueness of the factorization follows from the uniqueness of the factorization in $F[x]$, for another factorization in $R[x]$ would be a factorization in $F[x]$. Indeed, to see this suppose that $f = gh$ with $g, h \in F[x]$. Then take contents. The content of f must be a unit for otherwise the principal part of f would have to be trivial, and f is not a constant. By the Gauss Lemma the contents of g and h are units. Then we are left with $f = \text{pp}(g) \text{pp}(h)$, an equation in $R[x]$ that makes either of the principal parts a unit of $R[x]$.

Then any factorization in $R[x]$ has the same number of factors, and any two factorizations are equal up to a fraction u , which is a unit of $F[x]$. We just need to make sure u is a unit of R . Write uniqueness of the factorization in $F[x]$ as: $f_1 = u f'_1$ and $f_i = f'_i$ for $i \neq 1$. It is clear that $u \in R$, since otherwise f_1 couldn't be a polynomial in $R[x]$. But since f_1 is irreducible in $R[x]$ and f'_1 is not a unit of $R[x]$, we conclude that u is a unit of $R[x]$. $\square$

Lemma 5.12. *In a UFD, irreducible elements generate prime ideals.*

Proof. Let f be irreducible. Suppose that $pq \in (f)$. Then $pq = fg$ for some $g \in R$. Then $p_1 \dots p_k q_1 \dots q_\ell = f g_1 \dots g_m$. But since f is irreducible and factorization is unique we conclude that f must be one of the p_i or one of the q_i . Then f divides p or q , i.e. either $p \in (f)$ or $q \in (f)$. $\square$

6. UNIQUE FACTORIZATION DOMAINS (ALTERNATIVE APPROACH)

Lemma 6.1. *$\text{pp}(f)$ is unique up to multiplication by unit.*

Proof. This follows from uniqueness up to units of gcd and lcm. $\square$

Definition 6.2. If $f \in F[x]$, define its *primitive part* to be a primitive polynomial $\text{pp}(f)$ such that $\text{cpp}(f) = df$ where

- $d \in R$ is the minimum common factor of the denominators of the coefficients of f , which may be defined as the product of the least elements among all the factors (in their unique factorization) of the denominators such that $df \in R[x]$.

- $c \in R$ is the greatest common divisor of the coefficients of df , which may be defined as the product of the greatest number of common elements in the factorizations of each of the coefficients in df .

Lemma 6.3 (Gauss). *Let R be a ring and F its field of fractions. A primitive polynomial $f \in R[x]$ is irreducible in $F[x]$ if and only if it is irreducible in $R[x]$.*

Proof. The direct implication is easy: suppose f is irreducible in $F[x]$ and let $f = gh$ for $g, h \in R[x]$. Then g, h are also in $F[x]$ so that either of them must be a unit of F , i.e. a fraction. But since they are polynomials in $R[x]$, then it must be an element of R . But since f is primitive we obtain a contradiction unless the number is a unit.

For the converse, suppose that f is irreducible in $R[x]$. To obtain a contradiction suppose that $f = gh$ in $F[x]$ with g, h not units in $F[x]$. Then $f = \text{cpp}(f) = \text{cpp}(g)\text{pp}(h)$ by the True Gauss lemma. Notice that we cannot take primitive parts of g and h separately since this would give perhaps rational contents. $\square$

Lemma 6.4. *If $f \in F[x]$ is irreducible (in $F[x]$), then its primitive part $\text{pp}(f)$ is irreducible in $R[x]$.*

Proof. Suppose that $\text{pp}(f) = gh$ for two non-unit polynomials $g, h \in F[x]$. Multiplying by c as in Definition 6.2 we obtain $\text{cpp}(f) = cgh \implies df = cgh \implies f = \frac{c}{d}gh$, a contradiction since f is irreducible in $F[x]$. This shows that $\text{pp}(f)$ is irreducible in $F[x]$. By Gauss lemma, since $\text{pp}(f)$ is irreducible in $F[x]$ and primitive, it must be irreducible in $R[x]$. $\square$

The following is also commonly referred to as the Gauss lemma:

Lemma 6.5. *If R is a UFD then $R[x]$ is a UFD.*

Proof. Let $f \in R[x]$. Let c be the greatest common divisor of the coefficients of f , so that $f = cf'$. Factor f' in $F[x]$ as $f' = f'_1 \dots f'_k$. Take primitive parts to write $f = c' \text{pp}(f'_1) \dots \text{pp}(f'_k)$. By Lemma 6.4 each of the $\text{pp}(f'_i)$ is primitive in $R[x]$. By the True Gauss lemma we get that $c' = c \in R$. Since R is a UFD we also obtain that c' may be factored into irreducible elements of R , which are also irreducible in $R[x]$.

Uniqueness of the factorization follows from uniqueness of the factorization in $F[x]$ and Gauss lemma: suppose that f may also be factored into irreducible elements of $R[x]$ as $g_1 \dots g_\ell$. We wish to show that this is a factorization in irreducible elements of $F[x]$, which we know to be unique up to multiplication by units of F . To show this we use the converse implication of Gauss lemma 6.3: after taking primitive parts to obtain $\text{pp}(g_i)$, which are irreducible and primitive elements of $R[x]$, which must be irreducible in $F[x]$. Then $i = \ell$, the constants coincide and $\text{pp}(g_i) = \text{pp}(f_i)$. Since taking primitive part is unique up to multiplication by unit, we are done.

By induction on the degree of $f \in R[x]$. Write $f = cf'$ for $c = \text{gcd}(\text{coefficients of } f)$. Then I claim that $f = cf'$ is a factorization of f into irreducible elements. f' is irreducible because it is linear and primitive; that is, writing $f' = c'f''$ is impossible because there is no way to factor a non-unit number from the coefficients of f' ; indeed, in such case, cc' would be a *larger* common factor of f . Here I define larger as follows using that R is UFD: if $a = a_1 \dots a_k$ and $b = b_1 \dots b_\ell$ then the gcd is the product of all common factors of a and b .

To conclude our induction now suppose that degree- n elements have factorization and let f be of degree $n + 1$. If f is irreducible we are done. Otherwise f can be expressed as the product of two positive degree elements, each of which is expressed in a product of irreducibles. $\square$

7. MODULES

It looks like most books (at least [?] and [?]) define a module to be an abelian group along with a certain operation with elements of the ring.

Which secretly just says that

Definition 7.1. Let R be a ring. An (*left*) R -module M is an abelian group such that there exists a (left) ring morphism

$$R \rightarrow \text{End}(M).$$

Indeed, the three usual requirements correspond to:

- (1) The endomorphism corresponding to $a \in R$ respect the group structure of the group M :

$$a(x + y) = ax + ay,$$

- (2) The representation map is a map of rings

$$(a + b)x = ax + bx, \quad (ab)x = a(bx).$$

8. ALGEBRAS

Definition 8.1. If R is a commutative ring, then a *commutative algebra* over R is a commutative ring S together with a ring morphism $R \rightarrow S$.

But actually I was expecting that S would be defined as a ring that is also an R -module. So let us note that a morphism of rings $R \rightarrow S$ gives a representation $R \rightarrow \text{End}(S)$ via left multiplication. But the other way around, given an endomorphism associated to some element in R , how do we assign an element of S so as to produce a map $R \rightarrow S$?

Now we briefly introduce finitely-generated algebras and modules. Upshot: a finitely-generated R -algebra S is such that $S \simeq R[x_1, \dots, x_n]/I$ for some ideal $I \subset R$. Another way of saying this is that there is a surjective morphism $R[x_1, \dots, x_n] \rightarrow S$. Finitely-presentedness is when $I = (s_1, \dots, s_k)$.

As an aside, recall that finitely presented k -algebras with no nilpotents are in correspondence with affine algebraic varieties.

Definition 8.2. Let $R \rightarrow S$ be a ring map.

- (1) We say $R \rightarrow S$ is of *finite type*, or that S is a *finite type R -algebra* if there exist an $n \in \mathbf{N}$ and a surjection of R -algebras $R[x_1, \dots, x_n] \rightarrow S$.
- (2) We say $R \rightarrow S$ is of *finite presentation* if there exist integers $n, m \in \mathbf{N}$ and polynomials $f_1, \dots, f_m \in R[x_1, \dots, x_n]$ and an isomorphism of R -algebras $R[x_1, \dots, x_n]/(f_1, \dots, f_m) \cong S$.

For modules, the definition of finitely generated R -module is that there exists a surjective morphism $R^n \twoheadrightarrow S$. So: finitely-generated algebra means elements are polynomials on the generators with coefficients in the ring, and finitely-generated module means elements are linear combinations on the generators with coefficients in the ring.

9. LOCALIZATION

The general recipe for localization is that $\mathcal{O}_U = \mathcal{O}_X[f^{-1}]$ where $U = X \setminus V(f)$. That is, localizing gives the sheaf of the variety without the zero set.

Definition 9.1. Let R be a ring and $S \subset R$ a multiplicative subset of R (containing 1). The *localization* of R at S is the ring RS^{-1} (where S^{-1} is just a set with the same elements as S but with a -1 exponent) with the operations of $(rs^{-1})(r_1s_1^{-1}) = rr_1s^{-1}s_1^{-1}$, $rs^{-1} + r_1s_1^{-1} = (rs_1 + r_1s)(s^{-1}s_1^{-1})$ and the identification given by $rs^{-1} = r_1s_1^{-1} \iff rs_1 = r_1s$. (Check this definition if R has zero divisors!)

In a simple case, consider a ring R and $f \in R$. The *localization of R at f* is $R[t]/(ft - 1) := R[f^{-1}] = R(f)$. The quotient basically says that t is the inverse of f .

Example 9.2. • Dyadic numbers: $\mathbb{Z}[2^{-1}]$, which are fractions of integers with denominators which are powers of 2.

• Laurent polynomials: $\mathbb{C}[t, t^{-1}]$.

Lemma 9.3. If $R[f^{-1}] = 0$ then f is nilpotent.

Proof. If $R[f^{-1}] = 0$ then $1 \in (ft - 1)$, which means that $1 = (1 - ft)p(t)$ for some $p(t) \in k[t]$. Then... $\square$

The localization a ring is not always a local ring. But if S is a prime ideal, then it is.

Exercise 9.4. Prove that localization of a Noetherian ring is Noetherian.

Proof. Let R be a Noetherian ring and $S \subset R$. Let $I_1 \subset I_2 \subset \dots$ be a chain of ideals in the localization $R[S^{-1}]$. Let $\pi : R \rightarrow R[S^{-1}]$ the canonical projection $r \mapsto r$.

Let I be any of the I_i . Consider $\pi^{-1}(I)$. This gives a chain of ideals in R , which must stabilize. To go back to the localization simply recall that $\pi^{-1}(I)[S^{-1}] = I$. $\square$

10. NOETHERIAN AND ARTINIAN RINGS

In this section we explain why can any affine algebraic variety be expressed as a union of irreducible affine algebraic varieties. Roughly, the proof is as follows. Hilbert's basis theorem assures that the ring of polynomials in n variables over a field is Noetherian, which guarantees that there are only finitely many prime ideals containing any given ideal. Since the radical of an ideal I is the intersection of all primes $\mathfrak{p}_i$ containing I , and the variety of the radical is the same as the variety of the ideal, we can express $V(\sqrt{I}) = V(\cap \mathfrak{p}_i) = \cup V(\mathfrak{p}_i)$.

Definition 10.1. A ring is called *Noetherian* if any increasing chain of ideals $I_1 \subset I_2 \subset \dots$ stabilizes. It is called *Artinian* if any descending chain of ideals $I_1 \supset I_2 \supset \dots$ stabilizes.

Exercise 10.2. A ring is Noetherian if and only if all its ideals are finitely generated.

Proof. First we prove that if not every ideal is finitely generated then R cannot be Noetherian. Indeed, if I is not finitely generated we construct an ascending chain by of ideals by starting with any generator of I and adding other generators one by one.

For the converse suppose that all ideals of R are finitely generated and pick an ascending chain $I_1 \subset I_2 \subset \dots$. Consider $I = \bigcup_i I_i$, which is an ideal since for any $a \in I$ there is i such that $a \in I_i$ so that $ra \in I_i$ for any $r \in R$. However, since it is finitely generated, all the generators have to be in some I_j , and thus the chain stabilizes. $\square$

Theorem 10.3 (Hilbert basis theorem). *If R is Noetherian then so is $R[x]$.*

Exercise 10.4. If A is Noetherian then so is $A[[t]]$.

Proof. Pass from an ideal in $A[[t]]$ to A by taking the coefficients of the monomials of minimal degree of each element in the ideal. Then take a finite set of generators. Then go back to $A[[t]]$ by taking the polynomials where the generators were taken. Then... $\square$

Exercise 10.5. The radical of an ideal is the intersection of all primes containing it, that is,

$$\sqrt{I} = \bigcap_{\mathfrak{p} \supset I} \mathfrak{p}_i.$$

Since in a Noetherian ring there are only finitely many primes containing any ideal, we obtain the irreducible decomposition of affine algebraic varieties.

11. TENSOR PRODUCT

Definition 11.1.

Theorem 11.2 (Universal property of tensor product). *Let M_1, M_2, M be modules over a ring R . For any bilinear map $B : M_1 \times M_2 \rightarrow M$ there exists a unique homomorphism $b : M_1 \otimes M_2 \rightarrow M$ making the following diagram commute*

$$\begin{array}{ccc} M_1 \times M_2 & \xrightarrow{\pi} & M_1 \otimes M_2 \\ & \searrow B & \downarrow b \\ & & M \end{array}$$

where $\pi(m_1, m_2) = m_1 \otimes m_2$.

Remark 11.3. For $R = k$ the above property shows that $\text{Bil}(M_1 \times M_2, k) = (M_1 \otimes M_2)^*$. If we assume that the vector spaces are finite dimensional, we can take duals to obtain

$$M_1 \otimes M_2 = (M_1 \otimes M_2)^{**} = \text{Bil}(M_1 \times M_2, k)^*,$$

which is how Whitney first defined tensor products.

Here's Misha's way to show that the universal property of tensor product implies its uniqueness. Consider the category $\mathcal{C}$ of pairs of an R -module M , and a bilinear map $M_1 \times M_2 \rightarrow M$. Then we prove that the tensor product $M_1 \otimes M_2$ is an initial object in such category, and it is easy to show that initial objects are unique.

Remark 11.4. Misha's alternative way of stating the universal property of tensor product is that there is a bijective correspondence

$$\mathrm{Hom}_R(M_1 \otimes_R M_2, M) \xleftrightarrow{\sim} \mathrm{Bil}(M_1 \times M_2, M).$$

The key observation is that the LHS are *linear* maps from a module to M while the RHS are *bilinear* maps.

Exercise 11.5 (Currying).

$$\mathrm{Hom}(M_1 \otimes_R M_2, M) = \mathrm{Hom}(M_1, \mathrm{Hom}(M_2, M))$$

Proof. This is implied by the universal property since the RHS is just $\mathrm{Bil}(M_1 \times M_2, M)$. $\square$

Exercise 11.6. Let

$$M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be exact. Show that

$$0 \longrightarrow \mathrm{Hom}(M_3, N) \xrightarrow{- \circ f_2} \mathrm{Hom}(M_2, N) \xrightarrow{- \circ f_1} \mathrm{Hom}(M_1, N)$$

is exact.

Proof. Injectivity of the first arrow is easy. For the middle term pick $f \in \mathrm{Hom}_R(M_2, N)$ in the kernel of $- \circ f_1$. To show that it is in the image of $- \circ f_2$ we look for $g \in \mathrm{Hom}_R(M_3, N)$ such that $f = g \circ f_2$.

Since $M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ is exact, we have an isomorphism $\bar{f}_2 : M_2/\mathrm{img} f_1 \xrightarrow{\sim} M_3$. Since $f \circ f_1 = 0$, we see that f induces a map $\bar{f}$ on the quotient since indeed, if $m_2 - m'_2 \in \mathrm{img} f_1$, we have $0 = f(m_2 - m'_2) = f(m_2) - f(m'_2)$.

Then for any $m_3 \in M_3$ we have a unique class $m_2 + M_1 \in M_2/\mathrm{img} f_1$. Thus we can define $g = \bar{f} \circ \bar{f}_2^{-1}$.

Note that this choice satisfies $f = g \circ f_2$. Indeed,

$$g \circ f_2(m_2) = \bar{f} \circ \bar{f}_2^{-1}(m_3) = \bar{f} \circ (m_2 + M_1) = f(m_2).$$

$\square$

Exercise 11.7. Let $R = \mathbb{Z}$. Find an example of an injective homomorphism $M_1 \rightarrow M_2$ such that $\mathrm{Hom}(M_2, N) \rightarrow \mathrm{Hom}(M_1, N)$ is not surjective.

Proof. Consider

$$\begin{array}{ccccc} 0 & \longrightarrow & 2\mathbb{Z} & \xrightarrow{i} & \mathbb{Z} \\ & & \downarrow \varphi & \nearrow \bar{\varphi} & \\ & & \mathbb{Z}/2\mathbb{Z} & & \end{array}$$

where

$$\begin{aligned} \varphi : 2\mathbb{Z} &\longrightarrow \mathbb{Z}/2\mathbb{Z} \\ 2n &\longmapsto [n]. \end{aligned}$$

Then there is no extension $\bar{\varphi} : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ making $\varphi = \bar{\varphi} \circ i$. Indeed, note that $\varphi(2) = [1]$, while $\bar{\varphi}(2 \cdot 1) = 2\bar{\varphi}(1) = [0]$. $\square$

The idea is that any element in the image must be divisible by any integer. This is why $\mathbb{Q}$ is an injective module according to the following definition. (A precise proof of why $\mathbb{Q}$ is injective can be found at 01D6.)

Definition 11.8. An R -module is *injective* if the functor $\text{Hom}(-, M)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{array}{ccccc} 0 & \longrightarrow & M_1 & \xhookrightarrow{\quad} & M_2 \\ & & \downarrow & \nearrow \exists & \\ & & N & & \end{array}$$

Exercise 11.9. Let

$$0 \longrightarrow M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be an exact sequence of R -modules. Prove that

$$0 \longrightarrow \text{Hom}_R(N, M_1) \xrightarrow{f_1 \circ -} \text{Hom}_R(N, M_2) \xrightarrow{f_2 \circ -} \text{Hom}_R(N, M_3)$$

is exact

Proof. Injectivity is simple. For the middle term let $f : N \rightarrow M_2$ be in the kernel of $f_2 \circ -$, that is, $f(N)$ is killed by f_2 .

We want to define a map $g : N \rightarrow M_1$ such that $f_1 \circ g = f$. But observe that the image of f already lies in $M_1 = \text{Ker } f_2$. Thus f is already a map $f : N \rightarrow M_1$, save a more precise notation. $\square$

Exercise 11.10. Let $R = \mathbb{Z}$. Find an epimorphism (surjective homomorphism) of R -modules $M_2 \rightarrow M_3$ such that the corresponding map

$$\text{Hom}_R(N, M_2) \rightarrow \text{Hom}_R(N, M_3)$$

is not surjective for some R -module N .

Proof. Now consider

$$\begin{array}{ccccc} & & \mathbb{Z} & & \\ & \nearrow \exists & \downarrow \pi & & \\ 2\mathbb{Z} & \xrightarrow{\varphi} & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 \end{array}$$

for the same φ as in Exercise 11.7. Then $\pi(1) = [1]$, while any extension mapping $1 \rightarrow 2n$ gives $\varphi(2n) = 2\varphi(n) = [0]$. $\square$

Definition 11.11. An R -module N is *projective* if the functor $\text{Hom}_R(N, -)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{array}{ccccc} & & N & & \\ & \nearrow \exists & \downarrow & & \\ M_2 & \twoheadrightarrow & M_3 & \longrightarrow & 0 \end{array}$$

Exercise 11.12. A finitely generated R -module N is projective if and only if it is a direct summand of a free R -module R^n , i.e. there is a direct sum decomposition $R^n = N \oplus N'$.

Proof. Consider $R^n \simeq \bigoplus_{i=1}^n a_i R$ where N is generated by $a_1, \dots, a_n$. Apply projectivity to obtain

$$\begin{array}{ccccc} & & N & & \\ & \swarrow \exists \psi & \downarrow \text{id} & & \\ R^n & \xrightarrow{\varphi} & N & \longrightarrow & 0 \end{array}$$

where

$$\begin{aligned} \varphi : R^n &\longrightarrow N \\ (r_1 a_1, \dots, r_n a_n) &\longmapsto \sum r_i a_i. \end{aligned}$$

This is a section of φ in the exact sequence

$$0 \longrightarrow \text{Ker } \varphi \longrightarrow R^n \begin{array}{c} \xrightarrow{\varphi} \\ \xleftarrow{\psi} \end{array} N \longrightarrow 0$$

which gives decomposition

$$R^n \simeq \text{Ker } \varphi \oplus \text{Im } \psi.$$

More explicitly, the intersection of those modules is clearly trivial while any element of R^n may be written as

$$x = \underbrace{(x - \psi\varphi(x))}_{\in \text{Ker } \varphi} + \underbrace{\psi\varphi(x)}_{\in \text{Im } \psi}.$$

For the converse we start from the diagram given in the definition of projective module, add N' to every term, arriving at $R^n \simeq N \oplus N'$. We observe that free modules are projective by constructing an extension using a basis of R^n , and then back to N by taking the action of the extension restricted to N . $\square$

The upshot about projective modules is precisely that they can't be direct summands of a free module. Thus $\mathbb{Z}_2$ is not projective, since it has torsion, and so it cannot be a summand of $\mathbb{Z}^n$.

Thus, we create a non-projective module by quotienting the ring by any ideal - an element of the ideal is nilpotent, i.e. is a torsion element.

Further, it looks like projective modules are flat (confirm this!)

In fact, currying (Exercise 11.5) is the essential ingredient for the proof that tensor product is left exact. Such result follows from two lemmas.

Lemma 11.13. *Let*

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be an exact sequence of R -modules. Then for any R -modules N and N' , the sequence

$$0 \longrightarrow \text{Hom}_R(M_3 \otimes N', N) \longrightarrow \text{Hom}_R(M_2 \otimes N', N) \longrightarrow \text{Hom}_R(M_1 \otimes N', N)$$

is exact.

Proof. Apply $\text{Hom}_R(-, N)$, then $\text{Hom}_R(N', -)$ and then currying. $\square$

There is an (almost) converse to Exercise 11.6:

Lemma 11.14. *Let*

$$E : \quad M_1 \xrightarrow{\mu} M_2 \xrightarrow{\rho} M_3 \longrightarrow 0$$

be a complex, i.e. $\rho\mu = 0$, such that

$$0 \longrightarrow \operatorname{Hom}_R(M_3, N) \xrightarrow{\rho_N} \operatorname{Hom}_R(M_2, N) \xrightarrow{\mu_N} \operatorname{Hom}_R(M_1, N)$$

is exact for any R -module N . Then E is also exact.

And this is what implies right-exactness of tensor product:

Theorem 11.15. *Let*

$$M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be exact. Then

$$M_1 \otimes_R M \longrightarrow M_2 \otimes_R M \longrightarrow M_3 \otimes_R M \longrightarrow 0$$

is exact.

Proof. This is just Lemma 11.13 and Lemma 11.14!!! □

Exercise 11.16. Let $R = \mathbb{C}[t]$. Find an exact sequence of R -modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

and an R -module N such that the corresponding map

$$M_1 \otimes_R N \rightarrow M_2 \otimes_R N$$

is not injective.

Proof. Suppose the given exact sequence is just the ideal exact sequence of a point in $\mathbb{A}^1$. Then tensoring by M means thickening the point. Not having injectivity after tensoring means that tensoring the ideal by M does not give the ideal of the resulting variety. □

And from this it follows as corollary

Proposition 11.17. *Let $I \subset R$ be an ideal of a ring. Then $M \otimes_R (R/I) = M/IM$.*

Proof. Take the exact sequence $0 \rightarrow I \rightarrow R \rightarrow R/I$ and tensor by M . □

Exercise 11.18. Let I, I' be distinct maximal ideals in a ring R . Then $R/I \otimes_R R/I' = 0$.

Proof. The key observation is that since I and I' are maximal we must have $I + I' = R$. This means that $1 = f + f'$ for some $f \in I$ and $f' \in I'$. Then $1(a + I) \otimes (b + I') = (fa + I) \otimes (b + I') + (a + I) \otimes (f'b + I') = 0$. □

Now we discuss tensor product in the category of rings over a ring.

Definition 11.19. Let A, B and C be rings, and $C \rightarrow A$ and $C \rightarrow B$ homomorphisms. Considering A and B as C -modules, we can define a product on $A \otimes_C B$ by

$$a \otimes b \cdot a' \otimes b' := aa' \otimes bb'.$$

By the universal property this indeed extends to a product.

Example 11.20.

$$\mathbb{C}[t_1, \dots, t_k] \otimes_{\mathbb{C}} \mathbb{C}[z_1, \dots, z_n].$$

Example 11.21. For any homomorphism $\varphi : C \rightarrow A$, the ring $A \otimes_C (C/I)$ is isomorphic to $\frac{A}{\varphi(I)A}$ by Proposition 11.17.

Recall that the spectrum of a ring $\text{Spec } R$ is the corresponding algebraic variety.

Proposition 11.22. *Let $f : X \rightarrow Y$ be a morphism of affine varieties and $f^* : \mathcal{O}_Y \rightarrow \mathcal{O}_X$ the corresponding morphism. Let $y \in Y$ and $\mathfrak{m}_y$ the corresponding maximal ideal. Then*

$$\text{Spec} \left(\left(\mathcal{O}_X \otimes_{\mathcal{O}_Y} \frac{\mathcal{O}_Y}{\mathfrak{m}_y} \right) / \text{nilradical} \right) \cong f^{-1}(y).$$

Remark 11.23. Notice that we can see $y \in Y$ as a subvariety. In fact, it is, like X , a variety *over* Y . Thus this expression for $f^{-1}(y)$ matches that of the fibered product in Proposition 11.31.

Lemma 11.24. *Let A, B be finitely generated reduced $\mathbb{C}$ -algebras. If $R := A \otimes_{\mathbb{C}} B$, and $N \subset R$ nilradical (the set of nilpotent elements of R). Then*

$$\text{Spec}(R/N) = \text{Spec}(A) \times \text{Spec}(B).$$

Now we discuss...

Lemma 11.25. *Let A be a finitely-generated $\mathbb{C}$ -algebra. The intersection of all the prime ideals of A is its nilradical.*

Exercise 11.26. Let A and B be finitely generated $\mathbb{C}$ -algebras, $B \rightarrow A$ a homomorphism and $\mathfrak{m} \subset B$ a maximal ideal. Then $A \otimes_B (B/\mathfrak{m})$ can contain nilpotents.

Proof. The geometric idea is to project a parabola to an axis. At the point of tangency we obtain a fat point, whose algebra is not reduced. $\square$

However, over $\mathbb{C}$ the situation is different:

Theorem 11.27. *Let A, B be finitely generated reduced $\mathbb{C}$ -algebras. Then $A \otimes_{\mathbb{C}} B$ is reduced.*

It follows that

Theorem 11.28.

$$\text{Spec}(A) \times \text{Spec}(B) = \text{Spec}(A \otimes_{\mathbb{C}} B).$$

Now we can take preimages!

Lemma 11.29. *Let $f : X \rightarrow Y$ be a morphism of affine algebraic varieties, $f^* : \mathcal{O}_Y \rightarrow \mathcal{O}_X$ the corresponding ring homomorphism, $Z \subset Y$ a subvariety and I_Z its ideal. Denote by R_1 the quotient ring*

$$(\mathcal{O}_X \otimes_{\mathcal{O}_Y} (\mathcal{O}_Y / I_Z)) / \text{nilradical} = (\mathcal{O}_X / f^*(I_Z)) / \text{nilradical}.$$

Then

$$\text{Spec}(R_1) = f^{-1}(Z).$$

Proof. Analogous to preimage of a point. $\square$

The slogan is: adding polynomials to an ideal is the same as taking tensor product. This operation is given by Exercise 11.18.

We can also consider the diagonal of a product:

Lemma 11.30. *Let M be an affine algebraic variety, its diagonal $\Delta \subset M \times M$ and $I \subset \mathcal{O}_M \otimes_{\mathbb{C}} \mathcal{O}_M$ the ideal generated by $r \otimes 1 - 1 \otimes r$ for all $r \in \mathcal{O}_M$. Then*

$$\mathcal{O}_M \otimes_{\mathbb{C}} \mathcal{O}_M / I.$$

And now we introduce fibered product!

Proposition 11.31. *Let $X \rightarrow M$ and $Y \rightarrow M$ be morphisms of algebraic varieties, $R = \mathcal{O}_X \otimes_{\mathcal{O}_M} \mathcal{O}_Y$, and R_1 the quotient of R by its nilradical. Then*

$$\text{Spec}(R_1) = X \times_M Y.$$

Proof.

$$X \times Y \xrightarrow{f \times g} Z \times Z,$$

and compute $(f \times g)^{-1}(\Delta)$. By Lemma [?] we must have

$$A \otimes_{\mathbb{C}} B / (r \otimes 1 - 1 \otimes r) = A \otimes_C B$$

where $A = \mathcal{O}_X$, $B = \mathcal{O}_Y$ and $C = \mathcal{O}_Z$. □

We shall not explain why, but coproduct in the category of reduced finitely generated algebras is direct sum.

12. ARTINIAN ALGEBRAS

The aim of this section is to prove that a commutative algebra over a field which is finite-dimensional as a vector space and has no nilpotents (i.e. is semisimple) is a direct sum of fields; namely Theorem 12.15. The key idea is that in a finite-dimensional algebra we can use Chinese Remainder Theorem applied to the irreducible decomposition of the minimal polynomial of a non-invertible element to construct an idempotent element. This equivalent to the claim that if there are no idempotents the algebra is a field. Then the decomposition into fields is just a decomposition of unity into indecomposable idempotents, each of which gives a summand eR with no idempotents, i.e. a field.

This decomposition is used to prove that the preimage of a point by a finite map of affine algebraic varieties is a finite set of points, Theorem 13.3. Moreover, it also used quite a lot in the development of Galois theory in Section 15.

In this section algebras are not assumed to have unity.

Definition 12.1. The *minimal polynomial* of an element v in a finitely-generated k -algebra R a monic polynomial $P \in R[t]$ of smallest possible degree having v as a root.

Lemma 12.2. *The subalgebra generated by v is isomorphic to $k[t]/(P)$.*

Proof. Consider the map $k[t] \rightarrow Rv$, whose kernel is (P) since P divides any polynomial vanishing in v . □

Example 12.3.

$$\frac{\mathbb{R}[t]}{t^2 + 1} \simeq \mathbb{C}$$

while

$$\frac{\mathbb{R}[t]}{t^2 = 1} \simeq \mathbb{R} \oplus \mathbb{R}.$$

Recall that $k[t]$ is a UFD. Also recall that P is an irreducible element of $k[t]$ if and only if $k[t]/(P)$ is a field. This follows since $k[t]/(P)$ cannot have zerodivisors if P is prime, which is equivalent to irreducible in UFDs. These extensions are called *primitive*. [They are extensions of k obtained by adding a root of P .]

Definition 12.4. A commutative, associative k -algebra R not necessarily with unit is called an *Artinian algebra* if it is a finite-dimensional vector space over k . It is called *semisimple* if it has no nonzero nilpotents.

A commutative algebra with unity over a field is called an *Artinian ring* if it is finite-dimensional as a vector space over k .

We will show that any semisimple Artinian algebra is a direct sum (direct sum in the category of algebras is the direct sum of the rings equipped with natural operations) of fields, and this decomposition is uniquely defined.

An idempotent element e in R , i.e. $e^2 = e$, gives us a canonical decomposition

$$eR \oplus (1 - e)R$$

since $1 - e$ is also idempotent.

Exercise 12.5. Let $v \in R$ be an element of a finite-dimensional algebra R over k . Let $k[v]$ be the subalgebra generated by $1, v, v^2, \dots$. Prove that $P(v) = 0$ for a unique monic polynomial P of degree the dimension of $k[v]$.

Proof. Division algorithm. □

The polynomial above is called the *minimal polynomial* of v .

Exercise 12.6. Let $v \in R$ be an element of an Artinian ring over k , and $P(t)$ its minimal polynomial. Prove that $k[v]$ is isomorphic to $k[t]/(P)$.

Proof. Evaluation map. □

Exercise 12.7. Let $R = k[t]/P$ where $P \in k[t]$ is a polynomial decomposing as a product $P = P_1 \dots P_n$ of coprime polynomials. Prove that there exists an isomorphism $R \rightarrow \bigoplus_i k[t]/P_i$.

Proof. The natural map $[F]_P \mapsto ([F]_{P_1}, \dots, [F]_{P_n})$ is well defined. It is surjective and injective by the Chinese remainder theorem. If $[F]_P = [t]_P$, i.e. $F - t = PQ = P_1 \dots P_n Q$, it's immediate that $[F]_{P_i} = [t]_{P_i}$ for all i . □

Exercise 12.8. Let R be a semisimple Artinian ring with all idempotents equal to 1 or 0. Prove it is a field.

Proof. Suppose R is not a field, thus there is an element x which is not invertible. Consider the algebra $k[x]$ generated by x . By a Exercise ?? we know that $k[x] \simeq k[t]/P$ where P is the minimal polynomial of x . Now decompose P in its unique irreducible factorization. Since there are no nilpotents in R , there are no nilpotents in $k[x] \simeq k[t]/P$, which implies that the irreducible factorization of P is square-free, which means that the factors of P are coprime and we can apply the Chinese Remainder theorem. That is, by Exercise 12.7, we know that $k[t]/P$ is a direct sum of fields $\bigoplus k[t]/P_i$. There is an idempotent element here: $(1, 0, \dots, 0)$. □

Two idempotents $e_1, e_2 \in R$ are called *orthogonal* if $e_1 e_2 = 0$. If e_2, e_3 are orthogonal idempotents, then $e_1 = e_2 + e_3$ is also an idempotent satisfying $e_2, e_3 \in e_1 R$ and $e_1 R = e_2 R \oplus e_3 R$. In characteristic $\neq 2$, for any three idempotents e_1, e_2, e_3 , if $e_1 =$

$e_2 + e_3$ then e_2, e_3 are orthogonal. An idempotent $e \in R$ is called *indecomposable* if there are no non-zero orthogonal idempotents such that $e = e_2 + e_3$.

Exercise 12.9. Let R be a semisimple Artinian algebra, and $e \in R$ a non-decomposable idempotent. Then eR is a field.

Proof. Using Exercise ???. If there is an idempotent er we can decompose e as a sum of orthogonal idempotents, $e = (e - er) + er$. $\square$

The next exercise show that there is an indecomposable-orthogonal- decomposition of unity in an Artinian ring:

Exercise 12.10. Let R be a semisimple Artinian ring over a field k , $\text{char } k \neq 2$. Prove that 1 can be decomposed to a sum of indecomposable orthogonal idempotents and such decomposition is unique.

Proof. If 1 is indecomposable we are done. If it is decomposable, say $1 = e_1 + e_2$ then we can write $R = e_1 R \oplus e_2 R$. This implies that $e_1 R$ and $e_2 R$ are vector subspaces of R of strictly less dimension, and thus we can repeat the process and finish in finitely many steps. For uniqueness we write $1 = e_1 + \dots + e_k = e'_1 + \dots + e'_k$. Multiplying by e_1 we obtain $e_1 = e_1 e'_1 + \dots + e_1 e'_k$, which, if the decompositions do not coincide, is a decomposition of e_1 into orthogonal idempotents, a contradiction. $\square$

Exercise 12.11. Let R be an Artinian ring over a field k , $\text{char } k \neq 2$. Then R is a direct sum of fields and this decomposition is unique.

Proof. This is just Exercises 12.10 and 12.8. $\square$

Exercise 12.12. Does it work in characteristic 2?

Proof. ChatGPT: should work but not this technique. Me: although we can decompose $R = eR \oplus (1 - e)R$, we cannot show this is a field unless we apply Chinese Remainder Theorem! Perhaps this was a mistake in the lectures, or else the whole orthogonality business is useless. $\square$

Exercise 12.13. Let R be an Artinian ring over a field k , $\text{char } k \neq 2$ and $1 = e_1 + \dots + e_n$ a decomposition of 1 to a sum of indecomposable orthogonal idempotents. Prove that R has precisely n prime ideals.

Proof. My idea is that the given decomposition of unity gives a decomposition of the algebra in fields, which we may just think as an affine algebraic variety which is just n points. Any prime ideal in the algebra is an irreducible subvariety of the n points, which in turn can only be a point, which is already one of the obvious n prime ideals. $\square$

The following result was proved in lectures:

Theorem 12.14. *An Artinian algebra without nilpotents contains an idempotent.*

The proof of the theorem constructs a maximal ideal $I \subset A$ which contains a unit, and thus is a field I .

Thus we obtain

Theorem 12.15. *An Artinian algebra with unity with no nilpotents is a direct sum of fields.*

Proof. We find an idempotent element e , so that we can decompose

$$A = eA \oplus (1 - e)A$$

where e is a unity on the first factor and so is $1 - e$ on the second factor, thus every component is a field [I think this might be wrong!!! In the home assignment we needed Chinese Remainder Theorem to get the fieldness; see Exercise 12.8] We repeat this process until we exhaust the dimension of A . $\square$

We can also prove this theorem for algebras without unity. We consider the kernel of the map $u_0 : a \mapsto au$ and put

$$A = uA \oplus \text{Ker } u_0 A.$$

Lemma 12.16. *Any decomposition of an algebra into a direct sum of fields is unique up to permutation of the factors.*

We devote the end of this section to explain the relationship between semisimplicity and the trace form. Namely, Theorem 12.22 states that a semisimple Artinian algebra has nondegenerate trace form. This is reminiscent of Cartan's theorem which says that a finite-dimensional Lie algebra is semisimple in the sense that can be decomposed as a direct sum of simple Lie algebras (i.e. without nontrivial proper ideals) if and only if the Killing form (i.e. trace of adjoint operators) is nondegenerate.

Definition 12.17. Let R be a k -algebra, and $g : R \times R \rightarrow k$ a symmetric k -bilinear form on R . g is called *invariant* if

$$g(x, yz) = g(xy, z).$$

If R has a unity we have

$$g(x, y) = g(xy, 1),$$

so that in fact g is determined by a linear functional $a \mapsto g(a, 1)$.

Example 12.18. In the ring $\mathbb{C}[x, y]/(x^{n+1}, y^{n+1})$ the form $\varepsilon(xy) = a_n n$ is nondegenerate.

Definition 12.19. Let R be an Artinian ring over k . The *trace form* is the bilinear form $a, b \mapsto \text{tr}(ab)$ where $\text{tr}(ab)$ is the trace of the left-multiplication by ab endomorphism, i.e. $L_{ab} \in \text{End}_k R$, $x \mapsto abx$.

Exercise 12.20. Let $[K : k]$ be a finite field extension in characteristic 0. Prove that the trace form is always non-degenerate.

Proof. Since $[K : k]$ is finite, we find that $K \simeq k^n$ and we can choose an orthonormal base e_i with respect to usual Euclidean norm on k^n , i.e. dot product. Then for any $a, b \in K$,

$$\text{tr}(ab) = \sum_i \langle abe_i, e_i \rangle$$

by definition. Thus, since K is a field there exist inverses and a unity, so that for any nonzero x , $\text{tr}_k(x, x^{-1}) = n \neq 0$. $\square$

Definition 12.21. A field extension is *separable* if the trace form is nonzero.

Theorem 12.22. *Let R be an Artinian algebra over k with nondegenerate trace form. Then R is semisimple. Conversely, for $\text{char } k = 0$, the trace form is nondegenerate for any semisimple Artinian ring.*

Proof. Suppose that $a \in R$ is a nilpotent element. We shall show that this prevents the trace form from being nondegenerate. Indeed, we observe that for any $x \in R$, the endomorphism ax has no eigenvalues. Suppose $ax \cdot y = \lambda y$ for some $y \in R$ and $\lambda \in k$. Then $(ax)^n \cdot y = \lambda^n y$ vanishes and thus $\lambda = 0$. Thus the trace of ax is zero.

For the converse we use that any semisimple Artinian ring is a direct sum of fields. Then the endomorphism ab acts on a vector $x \in R$ by $(a_1 b_1 x_1, \dots, a_n b_n x_n)$ by definition of direct sum of algebras. Thus the trace is the sum of the traces in each of the fields that compose R . By Exercise 12.20 we know that each trace form is non degenerate. $\square$

Note that the proof above uses commutativity of the product in R . The result also holds in noncommutative algebras, but the proof must be reformulated using the Jacobson ideal, see Definition 17.2.

Lemma 12.23. *The trace in the tensor product is the tensor product in of the traces.*

13. FINITE MORPHISMS

Remark 13.1. If M is a finitely generated R -module and $R \rightarrow R'$ is a ring morphism, then $M \otimes_R R'$ is a finitely-generated R' -module generated by the same generators of M .

Definition 13.2. A *finite morphism* of affine varieties is called *finite* if the $\mathcal{O}_X$ is a finitely generated $\mathcal{O}_Y$ -module. In this case, $\mathcal{O}_X$ is called an *integral extension* of $\mathcal{O}_Y$.

Theorem 13.3. *Let $f : X \rightarrow Y$ be a finite morphism of affine varieties. Then for any point $y \in Y$, the preimage $f^{-1}(y)$ is finite.*

Proof. Let R be the $\mathcal{O}_Y$ -module $R := \mathcal{O}_X \otimes_{\mathcal{O}_Y} \mathcal{O}_Y/\mathfrak{m}_y$, which is the coordinate ring of $f^{-1}(y)$ by Proposition 11.22 or Lemma 11.29.

Further, it is finitely generated because $\mathcal{O}_X$ is finitely generated over $\mathcal{O}_Y$ (Remark 13.1).

Now observe that R is an Artinian algebra: elements of $\mathcal{O}_X$ are finite $\mathcal{O}_Y$ -linear combinations of certain elements $f_1, \dots, f_n \in \mathcal{O}_X$, but in R we have tensored by $\mathcal{O}_Y/\mathfrak{m}_y$, so by definition of tensor product the $\mathcal{O}_Y$ -coefficients pass to the $\mathcal{O}_Y/\mathfrak{m}_y$ slot, where they collapse to complex numbers; thus R is a finite-dimensional vector space over $\mathbb{C}$. By Theorem 12.15, $\text{Spec}(R)$ is a finite set! $\square$

14. FIELD EXTENSIONS

We collect the basic language of field extensions here. It will be used later both for Galois theory and for the function fields which appear in Noether normalization.

Definition 14.1. A *field extension* of the field k is another field K which contains k . This is denoted $[K : k]$.

Definition 14.2. Let $[K : k]$ be a field extension. An element $x \in K$ is *algebraic* if it is a root of a nonzero polynomial with coefficients in k , and otherwise it is *transcendental*.

Definition 14.3. Let $[K : k]$ be a field extension. We say that $[K : k]$ is a *finite extension* if K is finite-dimensional as a vector space over k . The dimension of this vector space is denoted $[K : k]$ and is called the *degree* of the extension.

Definition 14.4. Let $[K : k]$ be a field extension. $[K : k]$ is *algebraic* if every element of K is algebraic over k .

Finite extensions are algebraic, but algebraic extensions need not be finite.

Definition 14.5. Let $[K : k]$ be a field extension, and let $a_1, \dots, a_m \in K$. We write $k(a_1, \dots, a_m)$ for the smallest subfield of K containing k and $a_1, \dots, a_m$. Equivalently, it is the fraction field of the k -subalgebra $k[a_1, \dots, a_m] \subset K$.

Definition 14.6. Let $[K : k]$ be a field extension. We say that $[K : k]$ is a *finitely generated field extension* if there exist elements $a_1, \dots, a_m \in K$ such that $K = k(a_1, \dots, a_m)$.

The simple upshot is that algebraic independence is the analogue of being transcendental, but for several elements at once.

Definition 14.7. Let $[K : k]$ be any field extension. $t_1, \dots, t_n \in K$ are called *algebraically independent* if there is no nonzero polynomial $P \in k[z_1, \dots, z_n]$ such that $P(t_1, \dots, t_n) = 0$.

Example 14.8. The numbers 1 and 2 are algebraically dependent over $\mathbb{Q}$. Indeed, the nonzero polynomial $P(z_1, z_2) = z_1 - 1$ satisfies $P(1, 2) = 0$. More generally, any two elements of $\mathbb{Q}$ are algebraically dependent over $\mathbb{Q}$.

Example 14.9. A single transcendental number, such as $\pi \in \mathbb{R}$, is algebraically independent over $\mathbb{Q}$. Indeed, algebraic independence of one element just says that there is no nonzero polynomial $P \in \mathbb{Q}[z]$ such that $P(\pi) = 0$. For two actual numbers, things get hard fast: for example e and $e^{\sqrt{2}}$ are algebraically independent over $\mathbb{Q}$, but this uses the Lindemann-Weierstrass theorem.

Example 14.10. The elements $x, y \in \mathbb{C}(x, y)$ are algebraically independent over $\mathbb{C}$. Indeed, there is no nonzero polynomial $P \in \mathbb{C}[z_1, z_2]$ such that $P(x, y) = 0$ as a rational function. In contrast, the elements $x, x^2 \in \mathbb{C}(x)$ are algebraically dependent, since $z_2 - z_1^2$ vanishes after setting $z_1 = x$ and $z_2 = x^2$.

Definition 14.11. Let $[K : k]$ be a field extension. A set $t_1, \dots, t_n \in K$ is called a *transcendence basis* of K over k if $t_1, \dots, t_n$ are algebraically independent over k and K is algebraic over $k(t_1, \dots, t_n)$.

Exercise 14.12. Let A be a finitely generated domain over a field k . Prove that $\text{Frac}(A)$ is a finitely generated field extension of k .

Exercise 14.13. Let $[K : k]$ be a finitely generated field extension. Assume that K is algebraic over k . Prove that $[K : k]$ is finite.

Proof. Suppose that $K = k(a_1, \dots, a_n)$. Let $K_0 = k$ and $K_i = K_{i-1}[a_i]$. Since K is algebraic over k , each a_i is algebraic over k , hence also over the larger field K_{i-1} . By Lemma 16.8, K_i is a finitely generated K_{i-1} -module. Moreover, K_i is a field: if $0 \neq h(a_i) \in K_{i-1}[a_i]$, let $m_i(t)$ be the minimal polynomial of a_i over K_{i-1} . Since $h(a_i) \neq 0$, the polynomial m_i does not divide h . Thus $\gcd(h, m_i) = 1$, and by Bacht-Bézout for polynomials over a field, Theorem 5.5, there exist $u(t), v(t) \in K_{i-1}[t]$ such that

$$u(t)h(t) + v(t)m_i(t) = 1.$$

Evaluating at a_i gives $u(a_i)h(a_i) = 1$, so $h(a_i)$ has inverse $u(a_i)$ in $K_{i-1}[a_i]$. Thus $K_i = K_{i-1}(a_i)$ is a finite extension of K_{i-1} . Inducting on i and multiplying degrees, $K = K_n$ is finite over k . $\square$

A field is called *algebraically closed* if all algebraic extensions are trivial. This is equivalent to the field containing the roots of all polynomials.

Lemma 14.14. *A field k is algebraically closed if and only if any polynomial $P \in k[t]$ is a product of linear factors.*

Proof. If P does not decompose, there is a polynomial of degree >1 dividing P , and the quotient $k[t]/\text{such polynomial}$ is an algebraic field extension. $\square$

There is a good notion of algebraic closure of a field obtained by adding roots and applying Zorn's lemma to ensure this process finishes. And it is unique.

15. GALOIS THEORY

Now we study finite extensions from the point of view of automorphisms. Geometrically, one should think of passing from a variety to a cover and then studying automorphisms of the corresponding extension of function fields.

In this section we aim to prove: let $[K : k]$ be a finite extension over with $\text{char } k = 0$. then there exists an element $l \in k$ generating k .

Theorem 15.1. *Let $\bar{k} : k$ be the algebraic closure of k , and $[K : k]$ a separable finite extension. Then*

$$K \otimes_k \bar{k} = \bigoplus \bar{k}.$$

Recall that by Galois theory we can see that there only finitely many field extensions. Namely Galois theory says that field extensions are in correspondence with subgroups of the Galois group, which is finite. But this a simpler case:

Lemma 15.2. *Let k be a field, and $A = \bigoplus_{i=1}^n k$. Then A contains only finitely many different k -algebras.*

Proof. $\square$

Now we can prove (without Galois theory):

Theorem 15.3. *let $[k : k]$ be a finite extension over with $\text{char } k = 0$. then there exists an element $l \in k$ generating k .*

Proof. Using finiteness of the contained algebras... And the following fun claim: if V_i are positive codimension subspaces of k^n , then $\bigcup V_i \neq k^n$. $\square$

Definition 15.4. Let $[K : k]$ be a finite extension. It is called a *Galois extension* if the algebra $K \otimes_k K$ is isomorphic to a direct sum of several copies of K .

Example 15.5. Let $K = k[t]/(P)$ be a primitive, separable extension with $\deg P = n$. Suppose that P has n roots in K . Then $[K : k]$ is Galois. Indeed, this is the Chinese remainder exercise once again:

$$K \otimes_k K = \frac{K[t]}{(P)} = \bigoplus_i \frac{K[t]}{(t - a_i)}.$$

Exercise 15.6. $K = k[t]/P$. $[K : k]$ is Galois if and only if P has n roots in $K[t]$.

Definition 15.7. Let $P \in k[t]$ be an irreducible polynomial. *Splitting field* is the minimal extension such that P factors into linear factors.

Equivalently, pick $P \in k[t]$ of degree n without multiple roots, and k of characteristic 0. Decompose P into irreducible factors and construct finite extensions $K_1 \subset \dots \subset K_\ell$ by adding the roots of each factor. This process ends (prove this) and the resulting field is the *splitting field*.

Splitting field is Galois by Exercise 15.6. We can think of the splitting field as embedded in $\bar{k}$. There is not a unique embedding but all embeddings are isomorphic.

Definition 15.8. If $[K : k]$ is a Galois extension, $\text{Aut}_k K$ is called *Galois group*.

Note that K^* acts on $K \otimes_k K$ by multiplication on the right and on the left.

Lemma 15.9. Let $[K : k]$ be a Galois extension. There is a bijection between the following sets:

- (1) $\text{Aut}_k K$.
- (2) Prime ideals in $K \otimes_k K$.
- (3) Homomorphisms $K \otimes_k K \rightarrow K$ which are linear with respect to the left action of K .

Proof. (2 $\implies$ 3) is simple. □

Lemma 15.10. Let $[K : k]$ be a Galois extension, and $K \otimes_k K = \bigoplus_{\nu \in \text{Aut}_k K} K_\nu$ the decomposition of $K \otimes_k K$ enumerated by elements of $\text{Aut}_k K$. Then the left μ_l and right μ_r actions of K^* on $K \otimes_k K$ are such that $\mu_l(a)e_\nu = \mu_r(\nu(a))e_\nu$.

Lemma 15.11. Let $[K : k]$ be a Galois extension, and $a \in K$ a $\text{Aut}_k K$ -invariant element. Then $a \in k$.

Lemma 15.12. Let $[K : k]$ be a Galois extension, and K' an intermediate field (which, as we will see in the proof, will also be Galois) Then

$$K' = K^{G'}$$

where $G' \subset \text{Aut}_{K'} K$ is the group of K' -linear automorphisms of K and $K^{G'}$ is the space of G' -invariants.

Theorem 15.13 (Main theorem of Galois theory). Let $[K : k]$ be a Galois extension. The subgroups $H \subset \text{Aut}_k K$ are in bijective correspondence with intermediate subfields $k \subset K^H \subset K$.

16. NORMAL VARIETIES

Recall that $\text{Spec}(R)$ is the set of all prime ideals of the ring R . We introduce a topology whose closed sets are the prime ideals contained in a given ideal.

Definition 16.1. A morphism of algebraic varieties $f : X \rightarrow Y$ is called *dominant* if its image is dense, i.e. Y is the Zariski closure of $f(X)$.

Proposition 16.2. A morphism of varieties $f : X \rightarrow Y$ is dominant if and only if $f^* : \mathcal{O}_Y \rightarrow \mathcal{O}_X$ is injective.

Proof. Suppose f^* is not injective and consider its kernel $\text{Ker } f^*$. Then the image of f is in the zero set of the kernel, $V(\text{Ker } f^*)$. Here we use that points of a variety are homomorphisms from the variety to $\mathbb{C}$, which follows from the Nullstellensatz. . . Dominantness means that the image is not contained in the zero set of an ideal. □

Lemma 16.3. *If $f : X \rightarrow Y$ is dominant and X is irreducible, then Y is irreducible. Moreover, $f^* : \mathcal{O}_Y \rightarrow \mathcal{O}_X$ can be extended to the fields of fractions $k(Y) \rightarrow k(X)$.*

Proof. Since f is dominant, f^* is injective, i.e. an embedding, irreducibility transforms to not having zerodivisors. Descending to the field of fraction also follows from not having zerodivisors. $\square$

Remark 16.4. All of a sudden we consider a divisor on the affine variety X given by $f \in \mathcal{O}_X$. Then $X \setminus D$ is also an affine variety given by the ideal $I + \langle ft - 1 \rangle \subset \mathbb{C}[x_1, \dots, x_n, t]$. Recall that the localization of a ring by an element is $\mathbb{C}/\langle 1 - ft \rangle$, where t is the inverse of f .

Definition 16.5. A morphism of irreducible algebraic varieties $f : X \rightarrow Y$ is called a *birational morphism* if the induced map on fraction fields $k(Y) \rightarrow k(X)$ is an isomorphism.

We further show:

Proposition 16.6. *Let $f : X \rightarrow Y$ be a birational morphism. Then there exists a divisor $Z \subset Y$ such that $f : (X \setminus f^{-1}(Z)) \rightarrow Y \setminus Z$ is an isomorphism.*

Now we introduce normal varieties.

Here's some notes from the past that may pop up in memory: The upshot is: let $\varphi : R \rightarrow S$ be a ring map. An element $s \in S$ is integral over R if there is a monic polynomial with coefficients in R that gives zero when you put s instead of the variable.

Integral closure is analogous to algebraic closure, but it is not the same construction. For a field, the algebraic closure is a larger field in which all algebraic equations split. For a ring, integral closure is relative to an ambient ring, usually the fraction field, and only adds elements satisfying monic polynomial equations.

Definition 16.7. Let $A \subset B$ be rings. An element $b \in B$ is called *integral over A* if the subring $A[b] = A \cdot \langle 1, b, b^2, \dots \rangle$ is generated by b and A , is finitely generated as an A -module.

Note that if A is finitely generated, $A \langle 1, b, b^2, \dots \rangle$ is obviously finitely generated as a ring, but not as a module, which is what we care about.

Lemma 16.8. *An element $x \in B$ is integral over $A \subset B$ if and only if the chain of submodules*

$$A \subset A \cdot \langle 1, x \rangle \subset A \cdot \langle 1, x, x^2 \rangle \subset \dots$$

terminates. Equivalently, x is the root of a monic polynomial with coefficients in A .

Proof. The chain statement is almost tautological. As for the second, express x^n as an element of the finitely-generated module. Conversely, if x is a root of a monic polynomial

$$x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0,$$

then x^n is an A -linear combination of $1, x, \dots, x^{n-1}$. Multiplying by powers of x and arguing by induction, every higher power of x is also an A -linear combination of $1, x, \dots, x^{n-1}$. Thus $A[x]$ is generated as an A -module by $1, x, \dots, x^{n-1}$. $\square$

Proposition 16.9. *Let R be an integrally closed domain and let $S \subset R$ be a multiplicative subset with $0 \notin S$. Then RS^{-1} is integrally closed.*

Proof. Let $x \in \text{Frac}(RS^{-1})$ be integral over RS^{-1} . Then there is a monic polynomial

$$x^n + c_{n-1}x^{n-1} + \cdots + c_0 = 0$$

with $c_i \in RS^{-1}$. Choose $s \in S$ so that $sc_i \in R$ for all i . Multiplying by the right power of s , we get

$$(sx)^n + sc_{n-1}(sx)^{n-1} + \cdots + s^n c_0 = 0.$$

This is a monic polynomial over R vanishing at sx . Since R is integrally closed, we have $sx \in R$. Therefore $x = (sx)/s \in RS^{-1}$. $\square$

Lemma 16.10. *In Noetherian rings, sums and products of integral elements is integral.*

Definition 16.11. An irreducible affine algebraic variety is called *normal* if its coordinate ring is integrally closed, that is, it coincides with its integral closure, which in turn is the set of all elements that are integral over its field of fractions.

Recall from [?, Exercise 3.17, Chapter 1] that for affine varieties this is equivalent to all the local rings are all integrally closed.

Proposition 16.12. *An affine variety X is normal if and only if for any finite, birational morphism $Y \rightarrow X$ is an isomorphism.*

Proof. They have the same fraction field, and $\mathcal{O}_X \subset \mathcal{O}_Y$ by dominant (birational maps are dominant), which ways that $\mathcal{O}_Y$ is finite over $\mathcal{O}_X$, which says that elements of $\mathcal{O}_Y$ are integral. Since X is normal then the elements of $\mathcal{O}_Y$ belong to $\mathcal{O}_X$. $\square$

Proposition 16.13. *A UFD is integrally closed.*

Proof. Let u, v be elements of the UFD A and suppose the $u/v \in k(A)$ is integral, i.e. there is a monic polynomial $P \in A[t]$ of degree n killing u/v . Then u^n is divisible by v . Write $v = p_1 \cdots p_k$. Then p_i divides u^n , which implies that it divides u . But then p_i divides both u and v , which we may take to be coprime, making p_i a unit for all i . $\square$

As a corollary, we see that taking product of an affine variety, whose coordinate ring is an UFD, with affine space, $X \times \mathbb{C}^n$ preserves normality since the resulting coordinate ring is $\mathcal{O}_X[x_1, \dots, x_n] = \mathcal{O}_X \otimes \mathbb{C}[x_1, \dots, x_n]$.

17. NAKAYAMA'S LEMMA

We would like to have an analogue of the inverse function theorem, i.e. a function that is surjective on the tangent bundle is locally invertible. However, this is not true. Consider B a locally free $\mathcal{O}_M$ -module.

The idea is to find local trivializations of projective modules, just as we have local trivializations of vector bundles.

Nakayama's lemma is sometimes also called Krull lemma. We motivate with the following question: for an ideal $I \subset A$, when is $\cap I^n = 0$? Answer: when

$$I \cdot \bigcap I^n = \bigcap I^n.$$

The setting is: $\mathfrak{a} \subset A$ an ideal, M finitely generated module. . .

Definition 17.1. Let A be a domain and M be an A -module. A *torsion* is the set of all $m \in M$ such that there exists $0 \neq a \in A$ such that $am = 0$.

Definition 17.2. If R is a ring, its *Jacobson ideal* is the intersection of all maximal ideals.

Lemma 17.3. $a \in R$ belongs to $J(R)$ if and only if for all $r \in R$, $1 + ar$ is invertible.

Definition 17.4. An ideal $\mathfrak{a} \subset R$ is *quasi-regular* if $1 + a$ is invertible for any $a \in \mathfrak{a}$.

Lemma 17.5. An ideal is quasi-regular if and only if it belongs to the Jacobson ideal.

Proof. Straightforward. $\square$

Recall that the intuition of localization is “to pass to a smaller set” - at least geometrically.

Definition 17.6. Let $S \subset A$ be a multiplicative system, and $A(S^{-1})$ the localization of A in A . The *localization* of M in S is $M(S^{-1}) := M \otimes_A A(S^{-1})$.

Lemma 17.7. The natural map $\tau : M \rightarrow M(S^{-1})$ has kernel $\{m \in M : ms = 0 \text{ for some } s \in S\}$.

Lemma 17.8 (Nakayama, first version). If $\mathfrak{a} \subset A$ is a quasiregular ideal and M is a finitely generated A -module. Then

$$M = \mathfrak{a}M \implies M = 0.$$

Lemma 17.9 (Nakayama). Let A be a domain and M a finitely generated A -module. If $M = \mathfrak{a}M$ for some nonzero ideal $\mathfrak{a} \subset A$ then $(1 + a)M = 0$ for some $a \in \mathfrak{a}$.

Lemma 17.10. Lemmas 17.8 and 17.9 are equivalent.

Proof. Converse is immediate, for forward implication uses Lemma 17.7. $\square$

Recall that we call a module *cyclic* when it is generated by a single element, i.e. $M = \langle x \rangle$. This gives a surjective map $A \rightarrow M$, $a \mapsto ax$ and so $M = A/\text{Ker}$. So cyclic modules are quotients of the ring by an ideal.

Proof. First prove it for cyclic M . The definition of quasiregular ideal and cyclicity of M give us $(1 - a)x = 0$ which means that $z = 0$ since $1 - a$ is invertible.

If M is generated by $z_1, \dots, z_n$, we have a matrix given by the equation $\mathfrak{a}M = M$. Namely

$$z_i = \sum a_{ij} z_j$$

then put

$$(1 - a_{11})z_1 = \sum_{i=2}^n a_{ij} z_j.$$

Dividing by $1 - a_{11}$ since it is invertible, we can “get rid of z_1 ”, and construct a submodule M_1 , namely the submodule generated by the remaining generators, and in fact it satisfies $\mathfrak{a}M_1 = M_1$. Apply induction hypothesis. $\square$

We won't prove the following theorem:

Theorem 17.11 (Cayley-Hamilton). Let M be a finitely-generated A -module. An A -linear map $\varphi : M \rightarrow M$ can be represented by a matrix, which has a characteristic polynomial P . Then $P(\varphi) = 0$.

The idea is to use this theorem to express the identity of M as a matrix with coefficients in A :

$$\begin{array}{ccc} A^n & \xrightarrow{\tilde{\varphi}} & A^n \\ \pi \downarrow & & \downarrow \pi \\ M & \xrightarrow{\varphi = \text{Id}_M} & M \end{array}$$

In fact, this almost concludes the proof of Nakayama's lemma from Atiyah-Macdonald.

Now we turn to applications. Recall that localization by a prime ideal is a local ring. This says that elements outside the maximal ideal are invertible. The converse is also true and is used along with Nakayama's lemma to prove the following result. The intuition behind this is that if a map of modules induces a surjection on local quotients, then it is invertible locally (inverse function theorem!).

In the following lemma I think of M_1, M_2 as their corresponding affine varieties over the local algebra $(A, \mathfrak{m})$, which I can think as the stalk of a point. Then $M_1/\mathfrak{m}M_1 = M_1 \otimes_A A/\mathfrak{m}$ by Exercise 11.18. But we also know by Proposition 11.22 that this is the ring of the fiber of the point corresponding to the maximal ideal. So the following lemma says: if the map is surjective on fibers of the point, then it is surjective on the stalk. It looks like for a global statement, i.e. when A is not local, we could check this for every maximal ideal and conclude that φ is a surjective morphism.

Lemma 17.12. *Let $(A, \mathfrak{m})$ be a local ring and M_1, M_2 be finitely generated A -modules. Let $\varphi : M_1 \rightarrow M_2$ and suppose it induces a surjective map $M_1/\mathfrak{m}M_1 \rightarrow M_2/\mathfrak{m}M_2$. Then φ is surjective.*

This easily implies

Lemma 17.13. *Let M be a finitely generated A -module, A local. Let $x_1, \dots, x_n \in M$ be generators of $M \bmod \mathfrak{m}$. Then $x_1, \dots, x_n$ generate M .*

Theorem 17.14. *A finite dominant map is surjective.*

Proof. $A = \mathcal{O}_Y$, $B = \mathcal{O}_X$, $\mathfrak{m}_y \in A$ maximal ideal of y . We need dominance to suppose that $B \subset A$.

Then $\mathfrak{m}_y B \neq B$ by Nakayama's lemma.

Now consider $f^{-1}(y) = \text{Spec}(B \otimes_A A/\mathfrak{m}_y) = \text{Spec}(B/\mathfrak{m}_y B) \neq \emptyset$. □

Recall that projective modules are summands of free modules. Vector bundles over affine varieties are the same as projective modules over the rings of functions! That's a result by Serre, but we won't prove it here.

Theorem 17.15. *A finitely generated projective module over a local ring is free.*

Proof. Consider a map

$$\begin{aligned} A^n &\longrightarrow P \\ (a_1, \dots, a_n) &\longmapsto \sum a_i p_i. \end{aligned}$$

It is surjective and thus has a section since P is projective.

In writing $A^n = P \oplus Q$ we kill Q when taking quotient by the maximal ideal since $\mathfrak{m}Q = Q$ by Nakayama. □

As a final application consider the Groethendick group $K_0(R)$ where the subindex 0 denotes that we consider only exact sequences of projective modules.

Proposition 17.16. *Let $(A, \mathfrak{m})$ be a local ring. Then $K_0(A) = \mathbb{Z}$.*

Proof. Map $P \in K_0(Q)$ to $\dim_{A/\mathfrak{m}} P/\mathfrak{m}P$. Since projective modules are free, this is a bijection. $\square$

18. NOETHER NORMALIZATION LEMMA

Normalization means taking integral closure in the field of rational functions. Noether normalization lemma is the key ingredient to prove the Finiteness of Integral Closure theorem, Theorem 18.12, which says that the integral closure is a finitely generated algebra. In turn, this is the key fact needed to define normalization of a variety: it's the spectrum of the integral closure!

We start discussing finite quotients.

Theorem 18.1 (Noether theorem for G -invariants). *Let R be a finitely generated ring over $\mathbb{C}$ and G a finite group acting on R by automorphisms. The ring R^G of G -invariants is finitely generated.*

Proof. A bit involved - past lecture. $\square$

Lemma 18.2. *Let R be a Noetherian ring without zero divisors, G a finite group acting by automorphisms on R , and R^G the ring of G -invariants. Then $\varphi : \text{Spec } R \rightarrow \text{Spec } R^G$ is a finite, dominant morphism.*

Proof. Dominance is immediate because the map on rings is clearly injective, which is our definition of dominant map. We need to show that R is finitely generated as a R^G -module. Recall that to show that the ring is finitely generated is equivalent to finding minimal polynomials. $\square$

Example 18.3. $\mathbb{C}^2 = \text{Spec } \mathbb{C}[x, y]$ under the antipodal action of $\mathbb{Z}_2$. We obtain $\mathbb{C}^2/\pm 1 \simeq \mathbb{C}[a, b, c]/(c^2 - ab)$ because invariant polynomials are even polynomials.

Another example is $\mathbb{C}/\mathbb{Z}_n$, which gives $\text{Spec } \mathbb{C}[t^n]$ which is actually $\mathbb{C}$ again.

Theorem 18.4. *The fibers of the natural map $\varphi : \text{Spec } R \rightarrow \text{Spec } R^G$ are the orbits of G .*

Proof. We prove that the orbit is a finite set of points by showing the corresponding algebra is a semisimple Artinian algebra. $\square$

Now we proceed on our way to proving Noether's normalization lemma.

Lemma 18.5. *Let $[k : \mathbb{C}]$ be an extension of $\mathbb{C}$. Then either z is transcendental and K is isomorphic to the field of rational functions, or z is algebraic and the extension is finite.*

Proof. A polynomial vanishing on the point gives us finiteness once again. $\square$

We will use the terminology of field extensions from Section 14. In particular, a finitely generated field extension does not mean that $K = k(t_1, \dots, t_n)$ for some transcendence basis $t_1, \dots, t_n$. Relations among generators are allowed. This explains why, if $X \subset \mathbb{C}^n$ is an affine variety, then $k(X) = \text{Frac}(\mathcal{O}_X)$ is a finitely generated field extension of $\mathbb{C}$: the coordinate classes $\bar{z}_1, \dots, \bar{z}_n$ generate $k(X)$ as a

field, even though they may satisfy the relations coming from the ideal of X . See Exercise 14.12.

Theorem 18.6. *Let $[K : k]$ be a finitely generated field extension, e.g. $[k(X) : \mathbb{C}]$ where X is an affine algebraic variety. Then K is a finite extension of $k(t_1, \dots, t_n)$ where $t_1, \dots, t_n$ is a transcendence basis.*

Proof. Choose generators $a_1, \dots, a_m$ such that $K = k(a_1, \dots, a_m)$. Among them choose a maximal algebraically independent subset, say $t_1, \dots, t_n$. By maximality, every remaining generator is algebraic over $k(t_1, \dots, t_n)$. Since there are only finitely many remaining generators, adjoining them gives an algebraic finitely generated extension of $k(t_1, \dots, t_n)$, hence a finite extension by Exercise 14.13. $\square$

Let us spell out the notation in geometric terms. If $X \subset \mathbb{C}^n$, then

$$\mathcal{O}_X = \mathbb{C}[z_1, \dots, z_n]/I(X),$$

and the symbols t_i below are really the classes of the ambient coordinate functions $z_i + I(X)$ in this quotient ring, viewed also as elements of the function field $k(X)$. If some of these elements are algebraically independent, then they can serve as genuine independent coordinates on a dense part of X ; geometrically this will say that the projection to those coordinates is dominant.

Proposition 18.7. *Let $X \subset \mathbb{C}^n$ be an affine irreducible variety. Let $\Pi_k : X \rightarrow \mathbb{C}^k$ be the projection to the first k coordinates. The following are equivalent:*

- (1) Π_k is dominant and the extension $[k(X) : k(t_1, \dots, t_k)]$ is finite.
- (2) $t_1, \dots, t_k$ is a transcendence basis in $k(X)$.

Proof. Dominance of Π_k means that $\mathbb{C}[t_1, \dots, t_k] \rightarrow \mathcal{O}_X$ is injective. This is equivalent to saying that $t_1, \dots, t_k$ satisfy no nonzero polynomial relation, i.e. they are algebraically independent. Passing to fraction fields gives $\mathbb{C}(t_1, \dots, t_k) \subset k(X)$, the field generated by these independent coordinates inside $k(X)$.

The remaining condition that $[k(X) : \mathbb{C}(t_1, \dots, t_k)]$ is finite is precisely the statement that $k(X)$ is algebraic over the field generated by these elements; since $k(X)$ is finitely generated because $\mathcal{O}_X$ is a finitely generated algebra, algebraic here implies finite. Thus $t_1, \dots, t_k$ is a transcendence basis inside $k(X)$. $\square$

Here is the density point more concretely. The image of Π_k is not dense exactly when it is contained in the zero set of some nonzero polynomial $P \in \mathbb{C}[z_1, \dots, z_k]$. But being contained in that zero set means $P(t_1, \dots, t_k) = 0$ on X , i.e. a polynomial relation among the classes $t_i = z_i + I(X)$. Thus no polynomial relation among the t_i means the projection has dense image.

We are looking for conditions where this projection is a finite map. The projection Π_{n-1} is finite if and only if $P(z_n) = 0$ in $\mathcal{O}_X$ for some monic polynomial $P(t) \in \mathbb{C}[z_1, \dots, z_{n-1}][t]$.

Lemma 18.8. *Let $X \subset \mathbb{C}^n$ be an irreducible affine variety and suppose that the coordinate classes $z_1, \dots, z_k \in k(X)$ form a transcendence basis. Then there exists a linear coordinate change $z'_i := z_i + \lambda_i z_n$ such that z_n is finite over $z'_1, \dots, z'_k$, that is, z_n is a root of a monic polynomial with coefficients in $\mathbb{C}[z'_1, \dots, z'_k]$.*

Proof. Since $z_1, \dots, z_k$ is a transcendence basis, z_n is algebraic over $\mathbb{C}(z_1, \dots, z_k)$. Thus there is a polynomial relation involving z_n and $z_1, \dots, z_k$. After replacing z_i by $z'_i = z_i + \lambda_i z_n$, choose the constants λ_i generically so that the highest-degree

part of this relation does not vanish in the z_n direction. Then the relation can be viewed as a polynomial in z_n whose leading coefficient is a nonzero constant. Dividing by this leading coefficient gives a monic polynomial for z_n with coefficients in $\mathbb{C}[z'_1, \dots, z'_k]$. This is exactly the statement that z_n is finite over $z'_1, \dots, z'_k$. $\square$

Finally we can prove Noether's normalization lemma. The finite map obtained in this theorem is the key ingredient to prove the finiteness of the integral closure of an algebra.

Lemma 18.9 (Noether's normalization). *Let $X \subset \mathbb{C}^n$ be an irreducible affine variety. Then there exists a linear coordinate change such that the projection of X to the first k coordinates gives a finite map $X \rightarrow \mathbb{C}^k$.*

As a corollary we may obtain a dominant map:

Lemma 18.10. *Let $X \subset \mathbb{C}^n$ be an irreducible affine variety, z_i coordinates on $\mathbb{C}^n$, and $z_1, \dots, z_k$ transcendence basis on $k(X)$. Then there exists a linear change of coordinates $z'_i = z_i + \sum_{j=1}^{n-k} \lambda_{j+k} z_{j+k}$ such that the projection to the first k variables is a finite, dominant morphism.*

Proof. The fact that $z_1, \dots, z_k$ is a transcendence basis says, by Proposition 18.7, that the projection to the first k coordinates is dominant and generically finite. Apply Lemma 18.8 successively to the remaining coordinates $z_{k+1}, \dots, z_n$. After a suitable linear change, each remaining coordinate becomes a root of a monic polynomial with coefficients in $\mathbb{C}[z'_1, \dots, z'_k]$. Thus all coordinate functions of X are integral over $\mathbb{C}[z'_1, \dots, z'_k]$. Since $\mathcal{O}_X$ is generated by these coordinate functions, this makes $\mathcal{O}_X$ finite over $\mathbb{C}[z'_1, \dots, z'_k]$, which is exactly the finiteness of the projection. $\square$

Now we introduce a version that applies to non-irreducible varieties — at the cost of losing dominance.

Proposition 18.11. *Let $X \subset \mathbb{C}^n$ be an affine subvariety and X_i its irreducible components. Denote by k the maximal transcendence degree for $k(X_i)$. Then there exists a linear coordinate change z'_i such that the projection to the first k variables is finite.*

Recall that the integral closure of A in K is the set of all elements such that $A[x]$ is a finitely generated A -module. The integral closure of A is its integral closure in its field of fractions $k(A)$.

Theorem 18.12 (Finiteness of Integral Closure). *Let A be an integrally closed Noetherian ring without zero divisors and $[K : k(A)]$ a finite extension of its field of fractions, and B the integral closure of A in K . Then B is finitely generated as an A -module.*

Proof. Here is the expected strategy.

- (1) Use Noether normalization to find a polynomial subring $R = \mathbb{C}[x_1, \dots, x_d] \subset A$ such that A is finite over R .
- (2) Reduce the problem from A to the polynomial ring R .
- (3) Observe that K is a finite extension of $\text{Frac}(R)$, since it is finite over $k(A) = \text{Frac}(A)$ and A is finite over R .
- (4) Prove that the integral closure of R in K is finite over R . This is the serious algebra step, usually proved using trace.

- (5) Conclude that the integral closure of A in K is contained in this finite R -module, hence is finite over R and therefore finitely generated over A . $\square$

Lemma 18.13. *Let B be a domain over $\mathbb{C}$. Assume that there is an injective ring morphism*

$$A = \mathbb{C}[x_1, \dots, x_k] \rightarrow B$$

such that B is finitely generated as an A -module. Then the integral closure $\tilde{B}$ of B is finitely generated as an A -module. In particular, $\tilde{B}$ is a finitely generated ring.

Proof. The ring A is factorial by Gauss' lemma, hence integrally closed. Since B is finite over A , the field extension $\text{Frac}(B)/\text{Frac}(A)$ is finite. Theorem 18.12 implies that the integral closure of A in $\text{Frac}(B)$ is a finitely generated A -module. The integral closure $\tilde{B}$ is contained in this ring. Since A is Noetherian, $\tilde{B}$ is a finite A -module as well. $\square$

Definition 18.14. Let X be an irreducible affine variety, and let $A = \mathcal{O}_X$. Let $\tilde{A}$ be the integral closure of A in its field of fractions. Assume that $\tilde{A}$ is a finitely generated $\mathbb{C}$ -algebra. Then

$$\tilde{X} = \text{Spec } \tilde{A}$$

is called the *normalization* of X . The inclusion $A \subset \tilde{A}$ gives a morphism $\tilde{X} \rightarrow X$.

The normalization map is birational to X , and it is finite.

Example 18.15. Let $C \subset \mathbb{A}^2$ be the curve defined by

$$x^2 = y^3.$$

Consider the parametrization

$$\varphi : \mathbb{C} \rightarrow C, \quad t \mapsto (t^3, t^2).$$

On coordinate rings this gives

$$\mathbb{C}[x, y]/(x^2 - y^3) \cong \mathbb{C}[t^2, t^3] \subset \mathbb{C}[t].$$

Thus $\mathcal{O}_C = \mathbb{C}[t^2, t^3]$ and its fraction field is $\mathbb{C}(t)$. The element t lies in this fraction field, since $t = x/y$, and it is integral over $\mathcal{O}_C$ because it satisfies

$$t^2 - y = 0.$$

But $t \notin \mathbb{C}[t^2, t^3]$, so C is not normal.

Moreover $\mathbb{C}[t]$ is integral over $\mathbb{C}[t^2, t^3]$, because it is generated by the integral element t . It is integrally closed, since it is a polynomial ring in one variable. Therefore the integral closure of $\mathcal{O}_C$ in $\mathbb{C}(t)$ is $\mathbb{C}[t]$. Thus the normalization of C is

$$\text{Spec } \mathbb{C}[t] = \mathbb{A}^1,$$

with normalization map φ .

Lemma 18.16. *Let $\nu : \tilde{X} \rightarrow X$ be the normalization of an irreducible affine variety. Then ν is finite and birational. For every finite birational morphism $\mu : Y \rightarrow X$, there is a unique morphism $\tilde{X} \rightarrow Y$ over X . In particular, if Y is normal, then Y is isomorphic to $\tilde{X}$ over X .*

Proof. Let $A = \mathcal{O}_X$ and $\tilde{A}$ be its integral closure in $k(X)$. Since $\tilde{A}$ is finite over A , the map ν is finite. Since A and $\tilde{A}$ have the same fraction field, ν is birational.

Let Y be finite and birational over X . Then $\mathcal{O}_Y$ is a finite A -module inside $k(X)$. Thus every element of $\mathcal{O}_Y$ is integral over A , so $\mathcal{O}_Y \subset \tilde{A}$. This inclusion gives the unique morphism $\tilde{X} \rightarrow Y$ over X .

If Y is normal, then $\mathcal{O}_Y$ is integrally closed in $k(X)$ and contains A . Hence it contains $\tilde{A}$, and so

$$\mathcal{O}_Y = \tilde{A},$$

which gives the desired isomorphism. $\square$

Equivalently, normalization is the initial object in the category whose objects are finite birational morphisms $\mu : X' \rightarrow X$, and whose morphisms are maps over X . If one restricts to normal sources X' , the preceding lemma says that all objects are canonically isomorphic to the normalization.

19. ALGEBRAIC DIMENSION

Definition 19.1. Let M be an irreducible affine variety. The *algebraic dimension* $a(M)$ is the transcendence degree of the field extension

$$[k(M) : \mathbb{C}].$$

Equivalently, if $k(M)$ is finite over $\mathbb{C}(t_1, \dots, t_n)$, then $a(M) = n$.

We shall prove that this definition agrees with Krull dimension. The idea is to cut M by a principal divisor, then cut this divisor by a principal divisor on it, and continue in this way until we get a point. The Krull dimension shall be the number of such operations.

Definition 19.2. Let X be an affine variety, and let $f \in \mathcal{O}_X$ be a regular function which does not vanish on any irreducible component of X . The zero set of f is called a *principal divisor* on X . Its irreducible components are called *divisors* on X .

Lemma 19.3. Any divisor in $\mathbb{A}^n$ is principal.

Proof. Let $I = (f) \subset \mathbb{C}[t_1, \dots, t_n]$ be a principal ideal. Write

$$f = \prod_i f_i^{\alpha_i}$$

as the prime decomposition of f . Since $\mathbb{C}[t_1, \dots, t_n]$ is factorial, the ideals (f_i) are prime. Moreover

$$V(f) = \bigcup_i V(f_i).$$

Since all (f_i) are prime, this is the irreducible decomposition of $V(f)$. Thus each irreducible component of $V(f)$ is the zero set of a single equation, hence is principal. $\square$

Lemma 19.4. Let D be an irreducible divisor in $\mathbb{A}^n$. Then

$$a(D) = n - 1.$$

Proof. By Lemma 19.3, we can write $D = V(f)$ for an irreducible polynomial $f \in \mathbb{C}[t_1, \dots, t_n]$. After reordering the coordinates, suppose that f depends nontrivially on t_n .

The equation $f = 0$ gives one algebraic relation among the coordinate functions on D , so

$$a(D) \leq n - 1.$$

On the other hand, the classes of $t_1, \dots, t_{n-1}$ in $\mathcal{O}_D$ are algebraically independent. Indeed, if a nonzero polynomial in $\mathbb{C}[t_1, \dots, t_{n-1}]$ vanished on D , it would be divisible by f , which is impossible because f depends on t_n .

Finally, the class of t_n in $k(D)$ is a root of the polynomial f , viewed as a polynomial in t_n with coefficients in $\mathbb{C}[t_1, \dots, t_{n-1}]$. Thus $k(D)$ is finite over $\mathbb{C}(t_1, \dots, t_{n-1})$, and hence $a(D) = n - 1$. $\square$

The point is that t_n satisfies one polynomial equation over $\mathbb{C}(t_1, \dots, t_{n-1})$. Thus, after passing to rational functions, adjoining t_n is an algebraic extension. Since the field is generated by this one algebraic element over $\mathbb{C}(t_1, \dots, t_{n-1})$, the extension is finite.

The length of a chain

$$A_1 \subset A_2 \subset \dots \subset A_n$$

is $n - 1$, namely the number of inclusions.

Definition 19.5. The *Krull dimension* of a ring A is the maximal possible length of a chain of prime ideals

$$0 \neq \mathfrak{p}_1 \subset \mathfrak{p}_2 \subset \dots \subset \mathfrak{p}_n \subset A.$$

Definition 19.6. The *Krull dimension* of a variety X is the maximal possible length of a chain of nonempty, irreducible, distinct subvarieties

$$X_1 \subset X_2 \subset \dots \subset X_n.$$

Theorem 19.7. Let X be an affine variety. Then the Krull dimension of X as a variety is equal to the Krull dimension of its coordinate ring $\mathcal{O}_X$.

Proof. Irreducible subvarieties of X correspond to prime ideals of $\mathcal{O}_X$ by taking vanishing ideals. This correspondence reverses inclusions: if $Y \subset Z$, then

$$I(Z) \subset I(Y).$$

Thus a chain of irreducible subvarieties of X is the same thing as a chain of prime ideals in $\mathcal{O}_X$, with the order reversed. The two maximal possible lengths are therefore equal. $\square$

We shall prove below that for any affine variety, the algebraic dimension and the Krull dimension coincide.

Lemma 19.8 (Nakayama for local rings). Let A be a Noetherian local ring, and let $\mathfrak{p}$ be its maximal ideal. Let M be a finitely generated A -module. If

$$\mathfrak{p}M = M,$$

then $M = 0$.

Definition 19.9. Let $A \subset B$ be rings. We say that B is a *finite extension* of A if B is finitely generated as an A -module.

If $A \subset B$ and $\mathfrak{q} \subset B$ is an ideal, the *contraction* of $\mathfrak{q}$ to A is the ideal

$$\mathfrak{q} \cap A \subset A.$$

It is just the set of elements of A which already lie in $\mathfrak{q}$.

Lemma 19.10 (Going down). *Let $A \subset B$ be domains, and assume that B is a finite extension of A . Let $\mathfrak{q} \subset B$ be a nonzero prime ideal. Then*

$$\mathfrak{p} := \mathfrak{q} \cap A$$

is a nonzero prime ideal of A .

Proof. The contraction of a prime ideal is prime, so it remains to prove that $\mathfrak{p} \neq 0$. Choose $0 \neq b \in \mathfrak{q}$. Since B is finite over A , the element b is integral over A . Thus b satisfies a monic equation

$$b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0$$

with $a_i \in A$. Choose such an equation with n minimal. Then $a_0 \neq 0$, otherwise we could divide by b because B is a domain. But

$$a_0 = -b(b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1)$$

lies in $\mathfrak{q}$. Hence $0 \neq a_0 \in \mathfrak{q} \cap A$. $\square$

Alternative proof. Let

$$S = A \setminus \{0\}.$$

Suppose that $\mathfrak{q} \cap A = 0$. Then $S \cap \mathfrak{q} = \emptyset$, so $\mathfrak{q}S^{-1}B$ is a nonzero proper ideal in $S^{-1}B$.

On the other hand, $S^{-1}A = \text{Frac}(A)$ is a field. Since B is finite over A , the ring $S^{-1}B$ is a finite-dimensional algebra over $\text{Frac}(A)$. In particular it is Artinian. It is also a domain, because B is a domain. By Theorem 12.15, an Artinian algebra without nilpotents is a direct sum of fields. Since $S^{-1}B$ is a domain, this direct sum has only one summand. Thus $S^{-1}B$ is a field.

But a field has no nonzero proper ideals, contradicting the existence of $\mathfrak{q}S^{-1}B$. Therefore $\mathfrak{q} \cap A \neq 0$. $\square$

The proof of the next lemma first localizes at $\mathfrak{p}$. Nakayama shows that the fiber over $\mathfrak{p}$ is nonzero. Prime ideals in this fiber correspond precisely to prime ideals of B lying over $\mathfrak{p}$. The fiber is a finite-dimensional algebra over a field, hence Artinian. Thus the Artinian algebra classification shows that it has only finitely many prime ideals.

Lemma 19.11 (Going up). *Let $A \subset B$ be rings, with A Noetherian, and assume that B is a finite extension of A . Let $\mathfrak{p} \subset A$ be a prime ideal. Then there exist finitely many prime ideals $\mathfrak{q} \subset B$ such that*

$$\mathfrak{q} \cap A = \mathfrak{p}.$$

Proof. Let

$$S = A \setminus \mathfrak{p}.$$

Localize:

$$A_{\mathfrak{p}} = S^{-1}A, \quad B_{\mathfrak{p}} = S^{-1}B.$$

The ring $A_{\mathfrak{p}}$ is local, with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. Also $B_{\mathfrak{p}}$ is finite as an $A_{\mathfrak{p}}$ -module.

By Nakayama's lemma, Lemma 19.8, we have

$$B_{\mathfrak{p}} \neq \mathfrak{p}B_{\mathfrak{p}}.$$

Hence

$$C := B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$$

is a nonzero finite-dimensional algebra over the field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$. In particular, C is Artinian. Every prime ideal of C contains its nilradical. After quotienting by the nilradical, the Artinian algebra decomposition, Theorem 12.15, identifies the reduced quotient with a finite direct sum of fields. Thus there are only finitely many prime ideals in C .

Let $\bar{\mathfrak{q}} \subset C$ be a prime ideal, and let $\tilde{\mathfrak{q}} \subset B_{\mathfrak{p}}$ be its preimage. Then $\tilde{\mathfrak{q}}$ is prime. Put

$$\mathfrak{q} := \tilde{\mathfrak{q}} \cap B.$$

Then $\mathfrak{q}$ is prime in B . Moreover $\tilde{\mathfrak{q}} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$, because the quotient $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is a field. Contracting back to A gives $\mathfrak{q} \cap A = \mathfrak{p}$.

Conversely, every prime of B lying over $\mathfrak{p}$ arises from a prime ideal of C in this way. Thus there are only finitely many such prime ideals $\mathfrak{q}$. $\square$

The Cohen-Seidenberg theorem is essentially the union of the going-up and going-down lemmas.

Theorem 19.12 (Cohen-Seidenberg). *Let $A \subset B$ be a finite extension of Noetherian rings. Let*

$$\mathfrak{q}_1 \subset \mathfrak{q}_2 \subset \cdots \subset \mathfrak{q}_n \subset B$$

be a chain of prime ideals, and put

$$\mathfrak{p}_i = \mathfrak{q}_i \cap A.$$

Then

$$\mathfrak{p}_1 \subset \mathfrak{p}_2 \subset \cdots \subset \mathfrak{p}_n \subset A$$

is a chain of distinct prime ideals.

Conversely, any chain of prime ideals

$$\mathfrak{p}_1 \subset \mathfrak{p}_2 \subset \cdots \subset \mathfrak{p}_n \subset A$$

is obtained in this way from a chain of prime ideals in B .

Proof. We prove (i). Suppose that for some i we have

$$\mathfrak{q}_{i-1} \subset \mathfrak{q}_i$$

but

$$\mathfrak{p}_{i-1} = \mathfrak{p}_i.$$

Replacing A by $A/\mathfrak{p}_{i-1}$ and B by $B/\mathfrak{q}_{i-1}$, we reduce to the case where $A \subset B$ are domains and $0 \neq \mathfrak{q}_i \subset B$ is a prime ideal. The Going down lemma, Lemma 19.10, then gives $\mathfrak{q}_i \cap A \neq 0$, a contradiction. Thus the contractions are distinct.

We prove (ii). Now let

$$\mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n$$

be a chain of prime ideals in A . By the Going up lemma, Lemma 19.11, there is a prime $\mathfrak{q}_1 \subset B$ lying over $\mathfrak{p}_1$. Assume by induction that

$$\mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_r$$

have been chosen over

$$\mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_r.$$

Apply the Going up lemma to the finite extension

$$A/\mathfrak{p}_r \subset B/\mathfrak{q}_r$$

and to the prime $\mathfrak{p}_{r+1}/\mathfrak{p}_r$. This gives a prime ideal $\mathfrak{q}_{r+1}/\mathfrak{q}_r$ of $B/\mathfrak{q}_r$ lying over it. Equivalently, we get a prime $\mathfrak{q}_{r+1} \subset B$ with

$$\mathfrak{q}_r \subset \mathfrak{q}_{r+1} \quad \text{and} \quad \mathfrak{q}_{r+1} \cap A = \mathfrak{p}_{r+1}.$$

This completes the induction. $\square$

Lemma 19.13. *Let $f : X \rightarrow Y$ be a finite dominant morphism of irreducible affine varieties. Then the Krull dimension of X is equal to the Krull dimension of Y .*

Proof. The morphism f gives an injective finite ring extension

$$\mathcal{O}_Y \subset \mathcal{O}_X.$$

Any chain of prime ideals in $\mathcal{O}_Y$ lifts to a chain in $\mathcal{O}_X$ by Theorem 19.12. Conversely, any chain of distinct prime ideals in $\mathcal{O}_X$ contracts to a chain of distinct prime ideals in $\mathcal{O}_Y$, again by Theorem 19.12. Therefore the maximal possible lengths of chains are equal. $\square$

Theorem 19.14. *For any affine variety X , its algebraic dimension $a(X)$ is equal to its Krull dimension.*

Proof. Let $d = a(X)$. By Noether normalization, Lemma 18.9, there exists a finite dominant morphism

$$X \rightarrow \mathbb{A}^d.$$

By Lemma 19.13, X and $\mathbb{A}^d$ have the same Krull dimension. Thus it remains to prove that

$$\dim \mathbb{A}^d = d.$$

We prove this by induction on d . For $d = 0$, affine space is a point. Assume the result for affine varieties of algebraic dimension $< d$. Let

$$0 = \mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots$$

be a maximal chain of prime ideals in

$$\mathbb{C}[t_1, \dots, t_d].$$

The prime ideal $\mathfrak{p}_1$ contains a nonzero polynomial g . Writing g as a product of irreducibles, and using that $\mathfrak{p}_1$ is prime, we find an irreducible polynomial $f \in \mathfrak{p}_1$. Since $\mathbb{C}[t_1, \dots, t_d]$ is a UFD, (f) is a prime ideal. Thus

$$0 \subset (f) \subset \mathfrak{p}_1.$$

By maximality of the chain, $\mathfrak{p}_1 = (f)$.

Let $D = V(f) \subset \mathbb{A}^d$. Then D is an irreducible divisor. Prime ideals containing (f) correspond to prime ideals of

$$\mathbb{C}[t_1, \dots, t_d]/(f) = \mathcal{O}_D.$$

Hence

$$\dim \mathbb{A}^d = 1 + \dim D.$$

By Lemma 19.4, we have $a(D) = d - 1$. By the induction hypothesis,

$$\dim D = d - 1.$$

Therefore

$$\dim \mathbb{A}^d = 1 + (d - 1) = d.$$

This proves the theorem. $\square$

We finish with Hilbert's Nullstellensatz. The proof below is algebraic: it uses Noether normalization instead of the usual complex-analytic trick. This is more appropriate here, because the analytic proof is special to $\mathbb{C}$, while the algebraic proof works over any algebraically closed field.

Theorem 19.15 (Hilbert's Nullstellensatz). *Let k be an algebraically closed field. Then every maximal ideal of $k[x_1, \dots, x_n]$ is of the form*

$$(x_1 - a_1, \dots, x_n - a_n)$$

for some point $a = (a_1, \dots, a_n) \in k^n$. Moreover, for every ideal $I \subset k[x_1, \dots, x_n]$, we have

$$I(V(I)) = \sqrt{I}.$$

Proof. Let $\mathfrak{m} \subset k[x_1, \dots, x_n]$ be a maximal ideal, and put

$$A = k[x_1, \dots, x_n]/\mathfrak{m}.$$

Then A is a field, finitely generated as a k -algebra. By Noether normalization, there is a polynomial subring

$$k[t_1, \dots, t_d] \subset A$$

such that A is finite over it. Since A is a field and is integral over $k[t_1, \dots, t_d]$, the ring $k[t_1, \dots, t_d]$ is a field. This forces $d = 0$. Therefore A is a finite algebraic extension of k . Since k is algebraically closed, $A = k$. Thus the quotient map

$$k[x_1, \dots, x_n] \rightarrow A = k$$

is evaluation at a point $a = (a_1, \dots, a_n)$, and

$$\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n).$$

Now let $I \subset k[x_1, \dots, x_n]$. The inclusion

$$\sqrt{I} \subset I(V(I))$$

is immediate. Conversely, let $f \notin \sqrt{I}$. Then the ideal

$$J = (I, 1 - tf) \subset k[x_1, \dots, x_n, t]$$

is proper. Indeed, if $1 \in J$, then after setting $t = 1/f$ we would get that a power of f lies in I . By the first part, J is contained in a maximal ideal corresponding to a point $(a, b) \in k^{n+1}$. Then $a \in V(I)$ and

$$1 - bf(a) = 0,$$

so $f(a) \neq 0$. Thus f does not vanish on $V(I)$, and $f \notin I(V(I))$. Therefore

$$I(V(I)) \subset \sqrt{I},$$

and the theorem follows. □